Memory and State Control for LLM-Driven NPC Dialogue: Scoped Dual-Tier Memory and Pre-Commit Invariant Validation under a Paired, Oracle-Audited Evaluation

Parth Mital, Bhumika Singh, Pranesh Gopal, and Kalyanaraman P

***Abstract*—Large language models (LLMs) let non-player characters (NPCs) hold open-ended conversations, but they forget facts established early in a session, propose world changes that break game rules, and cost more as context grows. We present OBSIDIAN FLASK, an NPC dialogue engine that separates generation from state control: a dual-tier memory (an 8-turn buffer plus a FAISS store whose retrieval is scoped to the current location and NPC, packed into one context budget) and an event-sourced world state in which every LLM-proposed mutation must pass a symbolic pre-commit validation barrier, followed by dialogue repair when a proposal is rejected. Taking Generative Agents as the base paper, we compare the engine with rolling-context, flat vector-memory, generic retrieval, scored memory-stream, forgetting-curve and full-history baselines at an equal memory budget, in a 2×2 memory-by-control factorial with seven ablations, on seven worlds of 8 to 128 locations, with four models from three families. Every model output is committed four ways and audited by an invariant oracle that shares no code with the validator, and the system is paired against its own earlier version on the same scenarios. With a self-hosted 7B backbone the system recalled 82.4 [78.0, 86.8]% of facts 20 or more turns old on held-out worlds, against 0.0% for a rolling context of equal budget and 61.1% for a Generative-Agents-style memory stream; its gains over flat vector memory (81.0%) and a forgetting-curve memory (78.7%) were not significant. Scored on facts older than the buffer, scoping raised MRR@6 from 0.727 to 0.901. On identical outputs, attack success was 6.1% with direct writes and 1.2% with the earlier validator, and no output committed a state violation or succeeded as an attack through the corrected barrier, which rejected 1.1% of lawful live requests. Snapshot-plus-suffix recovery made all 300 randomised crashes consistent, against 77 before. Dialogue repair did not reduce judged contradictions. The results are preliminary: the run used the second of eight planned sample sizes and automatic scoring.**

***Index Terms*—Large language models, non-player characters, long-term memory, retrieval-augmented generation, event sourcing, runtime verification, interactive fiction, evaluation methodology.**

I. INTRODUCTION

TEXT-centric games mediate navigation, object interaction and character dialogue entirely through language. Unlike graphical engines, where physics enforces spatial constraints, a text world can only stay consistent through an explicit symbolic state. Games have traditionally driven NPCs with dialogue trees and finite-state machines, which obey the rules at negligible cost but restrict players to predefined choices. LLMs remove that restriction, and surveys of LLMs in games identify the coupling of generated text to game mechanics and state as an open problem [1].

Three failure modes stand in the way. First, *long-horizon forgetting*: facts that fall out of a bounded context window are lost, and facts inside a long context are used unevenly [2]; very long-term conversational memory remains weak even with retrieval [3], [4]. Second, *invariant violations*: LLMs are unreliable simulators of state transitions [5], [6] and lose narrative commitments over long interactions [7]; left unchecked they move players between unconnected rooms, invent items and grant unearned money. Third, *cost growth*: appending full history, or running several reflective LLM calls per turn [8], [9], raises tokens and latency per turn.

This paper evaluates an architecture that addresses all three by separating generation from state control (Section IV), and compares it with the base paper, Generative Agents [8] (Section II-B). Its components are individually standard: a turn buffer, dense retrieval, schema-constrained output, a rule validator, an append-only event log and a deterministic reducer. Our claim is therefore not that any component is new. The contribution is an evaluation that isolates what each mechanism buys, against independent baselines, with the costs it imposes measured rather than assumed.

A. Research questions

RQ1 (memory): does scoped dual-tier memory improve long-range recall per prompt token over rolling-context, vector-memory and generic retrieval agents? **RQ2** (state control): does a pre-commit validation barrier over event-sourced state reduce committed invariant violations, and at what cost in false rejections and dialogue–state desynchronisation? **RQ3** (separability): do memory and control contribute independently? **RQ4** (efficiency): what latency and tokens does each component add? **RQ5** (generalisation): do the effects hold across worlds of different authorship, genre and scale, and across backbones?

B. Contributions

1) A formal account of the engine as implemented: the world state, the event reducer and its replay and snapshot properties, the scoped retrieval operator with its post-filter limitation, and the validation predicate, including the coverage gaps found in the code (Sections IV-A and IV-B).

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Source code, benchmark and run outputs: https://github.com/parthmital/RAG-Driven-NPC-Narrative-Engine.

2) A *paired four-way commit protocol* with an *independent oracle*: each LLM output is committed through the corrected barrier, through the earlier barrier, through the reducer alone and by direct writes, and the four post-states are audited by an eight-invariant oracle that imports nothing from the validator. Common random numbers make every control comparison exactly paired, and lawful *twins* of every attack measure false rejection (Section IV-D).
3) A multi-world benchmark (four families, seven worlds, long-horizon sessions of 50 to 200 turns, seven attack categories) and a single-notebook harness that reports recall, retrieval quality, violations, false rejection, desynchronisation, latency, tokens and replay consistency together, with cluster-bootstrap intervals and pre-registered hypotheses (Sections V and VIII-B).
4) Empirical findings from three runs: the defects that a paired, oracle-audited evaluation exposed in the first version (unbounded currency gains, a load path that discards the logged suffix, dialogue that contradicts rejected changes), a paired before-and-after measurement of their corrections, and the hypotheses that the corrected system does not support (Sections IX and X).

The paper follows the order of the project report: the literature survey (Section II) and the research gap it leaves (Section III), the proposed methodology (Section IV), the dataset (Section V), the novelty claimed (Section VI), the algorithms (Section VII), the implementation (Section VIII) and the comparison of results, including the comparison with the base paper (Section IX).

II. Literature Survey

A. Grounded dialogue and text games

LIGHT conditions dialogue on rooms, objects and personas in a crowd-written fantasy world [10], and goal-driven agents in LIGHT were trained with reinforcement learning over factorised speech and action spaces [11]. Neither keeps a persistent, validated world state over long sessions. TextWorld [12] and Jericho [13] provide formally specified or human-authored games whose engines enforce rules, but they parse a constrained command language and have no open NPC dialogue. Callison-Burch et al. [14] framed Dungeons and Dragons as a dialogue challenge including state prediction, and STORIUM anchors collaborative stories with structured cards [15].

B. Base paper: Generative Agents

We take Generative Agents [8] as the base paper: it is the most direct prior architecture for LLM-driven characters with long-term memory in a game-like world. Twenty-five agents live in a sandbox town whose areas and objects form a tree. Each agent keeps a *memory stream* of natural-language observations. Retrieval scores every memory as the equally weighted sum of three min-max-normalised terms: recency (exponential decay with factor 0.995 per sandbox hour since last access), importance (an integer from 1 to 10 that the LLM assigns when the memory is created) and relevance (cosine similarity of embeddings to a query); the top-ranked memories that fit the context window enter the prompt. Reflections, higher-level memories that the LLM synthesises when the summed importance of recent events exceeds 150, and plans are stored in the same stream. Agents act through natural-language statements that the engine maps onto the sandbox. The evaluation measures believability: 100 human evaluators ranked interview answers from the full architecture, three memory ablations and a crowdworker condition, and a two-day end-to-end simulation measured information diffusion, relationship formation and coordination.

Three properties of the base paper motivate this work. First, memory is not bound to a validated symbolic state: the paper describes no check of the agents' statements against game rules, and it reports embellished memories as a failure mode. Second, memory is expensive: every memory write costs an LLM call for importance, reflection costs more, and the authors report that simulating 25 agents for two days cost thousands of dollars in token credits. Third, the evaluation ranks believability rather than measuring recall of established facts or rule violations, and memory designs are not compared at a matched token budget. Section IX-K compares the two systems.

C. Memory for LLM agents

Reflexion stores verbal self-reflections across trials [9]. In multi-session chat, models that summarise and recall earlier sessions outperform standard encoder–decoders [16]. Later systems add forgetting curves [17], knowledge-graph indexing [18], self-organising linked notes [19], multi-granularity summaries with retrieval refined by reinforcement learning [20], and compression of interactions into compact indexed memory units [21]. Benchmarks show that long-term memory is far from solved [3], [4], and surveys organise the design space [22], [23]. These systems store natural-language memories and most spend extra LLM calls per write; none binds memory to a validated symbolic state, and none is compared at a matched token budget in a game setting.

D. Retrieval and long context

Retrieval-augmented generation conditions generation on passages from a dense index [24]; generic RAG ranks by similarity alone and has no notion of a turn's spatial or social scope. Extending the context is not a substitute: models use the middle of long contexts poorly [2]. ReAct keeps a growing scratchpad of observations and actions [25]. Scoping by metadata is a filtered nearest-neighbour query; post-filtering a global top-k is known to lose in-scope neighbours, which motivates filter-aware indices [26]. We analyse exactly this effect in the engine's retrieval (Proposition 2).

E. State tracking and verifier-gated generation

LLMs track entities imperfectly [6] and are unreliable text-world simulators [5]. The LLM-Modulo position argues that LLM proposals should be checked by external model-based verifiers [27]; grammar-constrained decoding guarantees syntactic well-formedness but not semantic legality [28]. Prompt

TABLE I
SURVEYED SYSTEMS AGAINST THE PROPERTIES STUDIED HERE. "INDEP. AUDIT" MEANS VIOLATIONS ARE SCORED BY A CHECKER THAT DOES NOT SHARE CODE WITH THE COMPONENT BEING EVALUATED. ENTRIES DESCRIBE THE CITED PAPERS' OWN EVALUATIONS.

Work	Memory	World state	Pre-commit check	Indep. audit	Replay
LIGHT [10]	episode context	text descriptions	action checks	–	–
D&D dialogue [14]	dialogue history	predicted from text	–	–	–
Generative Agents [8] (base)	scored memory stream	sandbox tree	–	–	–
MemoryBank [17]	forgetting-curve store	none	–	–	–
A-Mem [19]	linked notes	none	–	–	–
TextWorld/Jericho	game engine	formal state	parser and rules	–	engine
PANGeA [31]	memory system	game level data	LLM judge	–	–
Phillips et al. [32]	dialogue history	goal states	LLM judge	–	–
NCP-Bench [7]	model context	fact ledger	–	LLM auditors	–
This work	scoped buffer + FAISS	event-sourced, typed	symbolic validator	yes (8 invariants)	yes

injection shows that model output cannot be trusted to obey instructions in its context [29]. Event sourcing, where state is derived from an append-only log of events, is used in industrial systems that need an auditable history [30].

F. LLM NPCs and narrative games, 2023 to 2026

PANGeA generates RPG content with a memory system and a validator in which the LLM itself judges player input against design rules [31]. Phillips et al. model conversations as puzzles whose goal-based states are advanced by an LLM that reads the dialogue history [32], so state transitions are again judged by a model. SceneCraft generates branching NPC scenes from authorial instructions and converts them into a game scripting language [33], and CALYPSO supports Dungeon Masters with LLM-generated text grounded in an established game setting [34]. NCP-Bench measures how interactive narratives drift under adversarial player interventions: the best model kept its commitments in 42% of runs after 20 turns, and fact-conflict rates ranged from 40% to 68% across models, as judged by LLM auditors [7].

G. Summary of the surveyed work

Table I contrasts representative systems on the properties this paper studies.

III. RESEARCH GAP

The survey leaves five gaps, which together define the problem this paper addresses.

RG1. **Memory is not bound to validated state.** From the base paper's memory stream to compressed or graph-indexed memories, memory systems store natural-language text; none ties what an NPC remembers to a symbolic world state that the model cannot corrupt (Table I).

RG2. **State changes are judged by a model or not at all.** In-game systems either let an LLM judge legality [31], [32] or apply model output directly; symbolic pre-commit validation of LLM-proposed mutations has not been evaluated in NPC dialogue.

RG3. **No independent audit.** Where violations are measured, they are scored by the component under test or by LLM auditors whose authors note that they can err on ambiguous text [7]; no evaluation pairs identical model outputs across commit paths and scores them with a rule checker that shares no code with the validator.

RG4. **No matched-cost comparison.** Memory designs are compared on accuracy, rarely at an equal prompt-token budget against a rolling context; the base paper reports cost only as thousands of dollars of token credits [8].

RG5. **Costs of control are unmeasured.** False rejection of lawful requests, dialogue that contradicts rejected changes, and replay and crash recovery are not reported together with recall and violations.

Five research questions follow from these gaps (Section I); Section IV gives the methodology that answers them and Section VI states what is new.

IV. PROPOSED METHODOLOGY

The methodology has two parts. The engine (Sections IV-A and IV-B) separates language generation from state control: the LLM proposes dialogue and state changes, a scoped dual-tier memory supplies context, and only proposals that pass a symbolic barrier reach an event-sourced state. The evaluation (Sections IV-C to IV-G) isolates each mechanism in a factorial design with ablations, pairs every control comparison on identical model outputs, and scores violations with an independent oracle. The data it runs on is described in Section V, the pseudocode in Section VII, and the implementation in Section VIII.

A. Formal model

The model below is the one the code implements; module paths refer to `Backend/` at the evaluated commit.

1) World state:

Definition 1 (World state)**.** A world state is a tuple $S = (\mathcal{L}, A, \mathcal{N}, \mathcal{O}, \pi, \iota, c, R, J, t)$ where $\mathcal{L}$ is the set of locations with directed adjacency $A \subseteq \mathcal{L} \times \mathcal{L}$ (the `connected_to` lists); $\mathcal{N}$ is the set of NPCs, each with a location, persona, knowledge, conditional secrets and an `alive` flag; $\mathcal{O}$ is the

set of objects, each with a location in $\mathcal{L} \cup \{\perp\}$ and an owner; $\pi \in \mathcal{L}$ is the player's location, ι the inventory and $c \in \mathbb{Z}_{\geq 0}$ the currency; $R : \mathcal{N} \times (\mathcal{N} \cup \{\text{player}\}) \to [-100, 100]$ is the trust map; J the journal; and t the turn (schemas/world_state.py).

The seed S_0 fixes the canonical entity sets $\mathcal{L}_0, \mathcal{N}_0, \mathcal{O}_0$. The active NPC and the player's identity are restored from `NPC_SPOKE` and `SESSION_START` events, so the whole stored state is derived from the log.

2) Events, reduction and replay: An event is $e = (id, t, \tau, p, ts)$ with type τ from 14 types (schemas/events.py) and payload p. The reducer δ (core/reducer.py) deep-copies the state and applies one event; it ignores moves to unknown locations, clamps trust to $[-100, 100]$, floors currency at zero, and ignores an NPC `location_id` change to a non-canonical location. State after T events is

$$S_T = \text{foldl}(\delta, S_0, [e_1, \ldots, e_T]), \qquad S_i = \delta(S_{i-1}, e_i). \quad (1)$$

Events are appended to SQLite with autoincrement ids, one transaction per turn, and every $\Delta = 16$ turns a snapshot (S_k, id_k) is written atomically, where id_k is the last event id (graph/definition.py).

Proposition 1 (Replay and snapshot equivalence)**.** *If (i) δ is a function of (S, e) only and (ii) every change to the hashed part of the live state is made by applying a logged event with δ, then for any snapshot taken after event k,*

$$\begin{aligned} \text{foldl}(\delta, S_0, e_{1..T}) &= \text{foldl}\big(\delta, \text{foldl}(\delta, S_0, e_{1..k}), e_{k+1..T}\big) \\ &= \text{foldl}(\delta, S_k, e_{k+1..T}). \end{aligned}$$

Proof. The first equality is the definition of a left fold over a concatenated list; the second substitutes the snapshot. Condition (i) holds in the code because the only time-dependent value the reducer writes, a journal entry's timestamp, is read from the event, which is persisted with it. In the Review-02 version, condition (ii) failed for the active NPC, which was set outside the log; it is now an event field. □

Recovery from the latest snapshot therefore costs $O(|S| + r)$ with r the number of events after it, against $O(|S| + T)$ for a full fold. The load path (core/recovery.py, `recover_state`) returns $\text{foldl}(\delta, S_k, e_{k+1..T})$, and a full fold when no usable snapshot exists or the snapshot is ahead of the log. The Review-02 load path returned S_k and discarded every event after the last snapshot unless a per-turn auto-save happened to succeed. Section IX-F measures both.

3) Proposals and the validation barrier: Each turn the LLM returns one JSON object validated by a Pydantic schema (schemas/llm_output.py): dialogue, narration, a one-sentence memory summary, a list of proposals $P_t = [(\tau_j, p_j)]$ and a list of proposed new entities. The barrier (game/validator.py) checks the proposals *in order against a running state*: with $\hat{S}_0 = S_t$, proposal j is accepted when $V(\tau_j, p_j, \hat{S}_{j-1}, S_t)$ holds, and then $\hat{S}_j = \delta(\hat{S}_{j-1}, e(\tau_j, p_j))$; a rejected proposal leaves $\hat{S}_j = \hat{S}_{j-1}$. The turn commits

$$\begin{aligned} E_t &= [\, e(\tau_j, p_j) \;:\; V(\tau_j, p_j, \hat{S}_{j-1}, S_t)\,], \\ S_{t+1} &= \text{foldl}(\delta, S_t, [e_{\text{speech}}] +\!\!+ E_t). \end{aligned} \quad (2)$$

V requires: a move's destination is canonical, adjacent to the current location and the turn's only move; a trust change names a canonical NPC and keeps that NPC's total change in the turn within ± 20; a taken object is canonical and lies at the player's location, and an NPC's drop or take checks possession or presence; an NPC move follows a connection; state and flag updates name a canonical target and a key; a journal entry has content; a currency delta is an integer, total spending in the turn does not exceed the opening balance $c(S_t)$, and the net gain in the turn does not exceed 50. Engine-only event types and any non-empty `new_entities` list are rejected. When a proposal is rejected, a repair node asks the model once to rewrite the reply so that it does not describe the rejected change, and replaces it with a neutral narration if the rewrite does not parse (graph/definition.py). The Review-02 barrier, which we evaluate as the "before" condition, checked every proposal independently against S_t, did not bound currency gains and did not repair the dialogue.

4) Invariants audited by the oracle: The evaluation's oracle (notebook section 4.1) re-derives facts from $(S_{\text{pre}}, S_{\text{post}}, \text{committed mutations})$ and checks eight invariants, written from the specification rather than the validator. With $\mathcal{L}_0, \mathcal{N}_0, \mathcal{O}_0$ from the seed and $\mathcal{A}$ the *undirected* closure of the seed adjacency:

O1 canonicity
post-state entity sets equal the seed's; the player, inventory and NPC locations reference canonical ids.

O2 conservation
each object is in exactly one place: a location, the inventory, or an NPC's possession.

O3 adjacency
$\pi_{\text{post}} = \pi_{\text{pre}}$ or $(\pi_{\text{pre}}, \pi_{\text{post}}) \in \mathcal{A}$.

O4 theft/duplication
a newly held object was at the player's location, on the floor or held by an NPC who is there, and is held once.

O5 relocation
objects move only to the player's location; NPCs move only along $\mathcal{A}$.

O6 trust bound
$|R_{\text{post}}(n, \text{player}) - R_{\text{pre}}(n, \text{player})| \leq 20$ for every NPC.

O7 solvency
total spending in the turn $\leq c_{\text{pre}}$ and $c_{\text{post}} \geq 0$.

O8 unearned currency
$c_{\text{post}} - c_{\text{pre}} \leq 50$.

O8 encodes a design bound that the Review-02 validator did not implement; the oracle found that gap in the Review-02 run, and the corrected validator now enforces it (Section IX-C).

5) Scoped dual-tier memory: Long-term memory is a set of entries $m_i = (\mathbf{v}_i, x_i, t_i, \ell_i, n_i)$: a unit vector, the text shown to the model, the turn, a location and an NPC. On commit, $\mathbf{v}_i$ is the embedding of the *player input* under all-MiniLM-L12-v2 [35], [36] ($d = 384$, L2-normalised), x_i is the model's memory summary (or the truncated input when absent), ℓ_i is the player's location after the turn's events and n_i the active NPC. The index is an exact inner-product FAISS index [37], so $\mathbf{q}^\top \mathbf{v}_i$ is cosine similarity. For query $\mathbf{q}$, current location ℓ

and NPC n, let $\mathrm{top}_m(\mathbf{q})$ be the m most similar entries in descending order and define the scope predicate

$$\phi_{\ell,n}(m_i) = \big(\ell_i = \varnothing \vee \ell_i = \ell\big) \wedge \big(n_i = \varnothing \vee n_i = n\big), \quad (3)$$

where empty metadata acts as a wildcard. Scoped retrieval with over-fetch factor $f = 5$ and threshold $\theta = 3$ is

$$\mathcal{R}(\mathbf{q}) = \begin{cases} \mathrm{first}_k\big[m \in \mathrm{top}_{\min(fk,|M|)}(\mathbf{q}) : \phi_{\ell,n}(m)\big] \\ \qquad \text{if this list has } \geq \theta \text{ entries}, \\ \mathrm{top}_k(\mathbf{q}) \quad \text{otherwise (relaxation fallback)}, \end{cases} \quad (4)$$

with $k = 6$. The fallback *replaces* the scoped list rather than extending it. The short-term tier is a ring buffer of the last $B_s = 8$ turns. When $|M| > 1000$ the store keeps its newest 500 entries. Memories and buffer share one budget of 1,600 characters (`pack_context` in llm/prompt_builder.py): the newest two turns are always kept, then retrieved memories in rank order, then older buffer turns, newest first, while the budget allows; a memory of a turn whose dialogue is already in the history is dropped as a duplicate.

Proposition 2 (Post-filter reachability)**.** *An in-scope entry m can be returned by the scoped branch of (4) only if its similarity rank among* all *entries is at most $fk = 30$. Hence scoped recall can fall below the recall of an exact filtered search whenever more than $fk - 1$ out-of-scope entries are more similar to $\mathbf{q}$ than m.*

Proof. The scoped branch filters $\mathrm{top}_{fk}(\mathbf{q})$; entries outside it are never inspected. □

The prompt contains a fixed system block that states the validator's limits from the same configuration constants, the canonical state of the current location and active NPC (with the NPC's secrets once its trust in the player reaches 60), player identity, the packed memories and history, and the query, capped at 16,384 characters (llm/prompt_builder.py).

B. Turn pipeline

Fig. 1 shows one turn. The player input is embedded and used to retrieve scoped memories, the prompt is assembled from the canonical state, the memories and the buffer, and the LLM returns one JSON object. Its proposals are parsed and checked by the barrier against a running state, the reply is repaired if a proposal was rejected, and only accepted proposals are appended to the event log and applied by the reducer; the commit node also updates the buffer, the FAISS store and, every 16 turns, the snapshot. Section VII gives the pseudocode and Section VIII the software that runs it.

C. Experimental conditions

All conditions share the backbone, decoding parameters, system prompt, canonical-state serialisation, output schema and player scripts; they differ only in memory and in how proposals reach state (Table II). Baselines are independent NumPy implementations that do not reuse the system's retrieval code, and every memory condition except B0, B1×2 and F4v1 is held to the same 1,600-character budget. Budget B for the rolling contexts is the mean token count of F4's memory and history blocks on the calibration probes, 389 tokens. Without a validator, schema-valid proposals are written straight into the state (direct writes), modelling an agent that writes state through a tool interface. B4 re-implements the base paper's retrieval rule without its importance term (which needs an LLM call per memory write), and B5 re-implements MemoryBank's forgetting curve [17]. F4v1 is the Review-02 system: the corrected reducer, event store and commit node, but the Review-02 validator, prompt builder and unbudgeted memory, and no repair.

D. Paired four-way commit protocol

Request seeds depend on the probe or turn and never on the condition, so identical prompts receive identical outputs (common random numbers). For probes, the *same* parsed output is then committed four ways (through the corrected barrier, through the Review-02 barrier, through the reducer alone, and by direct writes) and audited by the oracle of Section IV-A4, so differences between modes are attributable to the commit path alone. Lawfulness of a proposal is judged by the oracle on a single-proposal transition, which gives false-rejection and validator-gap rates that do not depend on the validator's own verdicts. Section VII-A gives the procedure.

E. Metrics

Let $y_j \in \{0, 1\}$ be correctness of recall probe j and r_j the rank of its gold memory among the retrieved list (undefined if absent).

$$\begin{aligned} \mathrm{Recall} &= \tfrac{1}{n}\textstyle\sum_j y_j, \\ \mathrm{MRR@}k &= \tfrac{1}{n}\textstyle\sum_j \mathbb{1}[r_j \leq k]\, r_j^{-1}, \\ \mathrm{Recall@}k &= \tfrac{1}{n}\textstyle\sum_j \mathbb{1}[r_j \leq k]. \end{aligned} \quad (5)$$

MRR@k and Recall@k are computed only on probes whose fact is older than the 8-turn buffer: the packing step drops a memory that duplicates a buffered turn, so a younger fact has no retrieval rank in F4 even when it is in the prompt. This rule was adopted after the first Review-03 run, which counted such probes as misses, and the reported run applies it as part of the notebook (Section VIII-E). For rolling conditions, the *gold-in-prompt* rate (the plant turn is in the history or retrieved list) replaces MRR. For probes, *attack success* is the share of attacks whose post-state violates an invariant (a secret leak for the secret category); *false rejection* is $\sum$ lawful twin proposals rejected $/ \sum$ lawful twin proposals; the *validator gap* is $\sum$ unlawful proposals accepted $/ \sum$ unlawful proposals over attacks and twins. The *violation rate* is

$$\mathrm{Viol}/100 = 100 \cdot \frac{\sum_{\text{turns}} \mathbb{1}[\text{post-state violates some invariant}]}{\sum_{\text{turns}} (\text{committed non-speech mutations})}. \quad (6)$$

The oracle audits the whole post-state, so a corruption that persists (for example a player left in a non-existent location) is flagged on every later turn, including turns that commit nothing; in live sessions the rate can therefore exceed 100. *Desynchronisation* is an automatic proxy: a turn is flagged when the barrier rejected a proposal but the dialogue has no

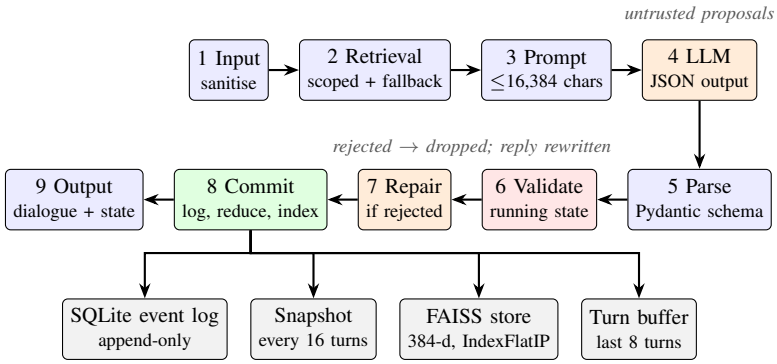


Fig. 1. Turn pipeline of OBSIDIAN FLASK (graph/definition.py). The LLM output is untrusted: proposals reach the world state only through the validation barrier (node 6) and the pure reducer inside the commit node (node 8); node 7 rewrites the reply when a proposal was rejected. The commit node writes the four stores; on later turns, retrieval (node 2) reads the FAISS store and prompt assembly (node 3) packs the memories and the turn buffer into one budget.

TABLE II
CONDITIONS. F1 TO F4 FORM THE 2×2 FACTORIAL; A1 TO A7 CHANGE ONE COMPONENT OF F4.

ID	Long-term memory	Prompt history	State control
F4	real `FAISSMemory` + real retrieval node (scope, fallback), packed into the budget	8-turn buffer, in the same budget	barrier + repair + event log + reducer (as shipped)
F4v1	as F4, unbudgeted	8-turn buffer	Review-02 barrier, no repair
B0	none	full history, oldest dropped only at the context limit	direct writes
B1 / B1×2	none	newest turns within B / $2B$ tokens	direct writes
B2	flat store of turn *summaries*, cosine rank, no scope	none	direct writes
B3	raw dialogue chunks + world lore documents, cosine rank	none	direct writes
B4	scored memory stream: relevance + recency (decay 0.995 per turn)	none	direct writes
B5	forgetting curve: relevance $\times \exp(-\Delta t/(S\tau))$, $\tau = 50$	none	direct writes
F1	none	rolling, budget B	direct writes
F2	none	rolling, budget B	barrier + repair + event log
F3	as F4	8-turn buffer	direct writes
A1	none	8-turn buffer	as F4
A2	as F4	none	as F4
A3	unscoped top-6	8-turn buffer	as F4
A4	scoped, no fallback ($\theta = 0$)	8-turn buffer	as F4
A5	as F4	8-turn buffer	schema-only events through the reducer
A6	as F4	8-turn buffer	validator, then in-place writes, no log
A7	as F4	8-turn buffer	as F4, recovery by full replay

refusal cue, or when, on a scripted move turn, the narration's claim of arrival disagrees with the committed move. A *contradiction judge*, Qwen2.5-7B-Instruct, a model of another family than the generator, labels whether a reply claims a change that was not committed. *Replay consistency* is the share of sessions whose state rebuilt from the log matches the live final state by a SHA-256 hash of the canonical JSON. A *crash trial* stops a commit at a random point (before, during or after the event write, or during or after the snapshot write) and compares the recovered state with the last committed one. Latency is reported as percentiles, and token cost as input and output tokens counted on the exact prompt ids and from vLLM usage fields.

F. Statistical analysis

Intervals are 95% cluster-bootstrap percentile intervals with 10,000 resamples [38], [39], clustering on session for recall and live metrics and on probe for probe metrics. Ratios are bootstrapped as ratios of resampled sums: for clusters g with numerator and denominator sums a_g, b_g and a resample $g_1^*, \dots, g_G^*$,

$$\hat{\rho} = \frac{\sum_g a_g}{\sum_g b_g}, \qquad \hat{\rho}^* = \frac{\sum_i a_{g_i^*}}{\sum_i b_{g_i^*}}. \tag{7}$$

Paired binary outcomes use the exact McNemar test [40], which conditions on the $n_{10} + n_{01}$ discordant pairs and tests $n_{10} \sim \mathrm{Bin}(n_{10} + n_{01}, \frac{1}{2})$; paired continuous outcomes use the Wilcoxon signed-rank test [41]. The confirmatory family (H1 to H3, six tests) is Holm-corrected [42]. The factorial is modelled on live turns as

$$\begin{aligned}\operatorname{logit} \Pr(y = 1) = \beta_0 + \beta_M M + \beta_C C \\ + \beta_{MC} MC + u_{\text{world}} + v_{\text{session}},\end{aligned} \tag{8}$$

with $M, C \in \{0, 1\}$ indicating dual-tier memory and the barrier and Gaussian random intercepts, fitted by variational Bayes (`BinomialBayesMixedGLM.fit_vb` in statsmodels [43]). We report posterior means and standard deviations and, as the notebook does, z = mean/sd with a two-sided normal tail probability. These are approximate posterior summaries, not frequentist Wald tests.

G. Hypotheses

Stated before the run, each against the strongest applicable baseline. A hypothesis is supported only if the corrected effect is significant with the predicted sign, or the stated threshold is met.

H1 F4 recall at $a \geq 20$ exceeds B1 at equal budget, and B2 and B3 at equal or lower token cost.

H2 Scoping plus fallback gives higher MRR@6 than unscoped flat retrieval (B2).

H3 The barrier reduces oracle-detected committed violations on probes and in live sessions.

H4 False rejection on lawful twins is at most 5% and validation adds under 5 ms at P95.

H5 Replay from the log reproduces the live state in 100% of event-sourced sessions.

H6 Memory mainly affects recall and control mainly affects violations, with small interaction.

H7 The directions of the H1 and H3 effects hold on W2, W3 and W4, and on two models of other families.

Four further tests compare the system with its Review-02 version on the same scenarios: **R1** the currency correction lowers attack success and the validator gap; **R2** snapshot-plus-suffix recovery makes crash recovery consistent; **R3** dialogue repair lowers contradictions; **R4** committed live violations fall. Each is confirmed if its Holm-corrected tests favour the correction, or if no failure remains after the correction.

V. Dataset Description and Usage

The dataset is a benchmark built for this study: seven worlds, teacher-forced recall sessions, adversarial probes with lawful twins, and live-session scripts, generated deterministically by the evaluation notebook from a master seed, together with the model outputs and metrics of the reported run. Section V-E lists the released files and how each is used.

A. World suite

A single hand-made world cannot support generalisation claims, and a benchmark written by the system's authors risks being tuned to it. The suite therefore has four families (Table III and Fig. 2): W1, the in-house dark-fantasy development world (Backend/game/world_seed.json); W2a to W2c, converted by a deterministic seeded script from the crowd-written LIGHT environment file [10] (CC BY-NC 4.0), with LIGHT's fragmented room graph joined by a seeded spanning tree and two templated secrets per NPC built from words that occur nowhere else in the world; W3, a 24-location science-fiction station authored in the notebook from a written specification before any system run; and W4-32 and W4-128, procedural graphs (random spanning tree plus about 30% extra edges) with templated names. No system setting was tuned on W2 to W4. W3 is held out from tuning but was *not* independently authored.

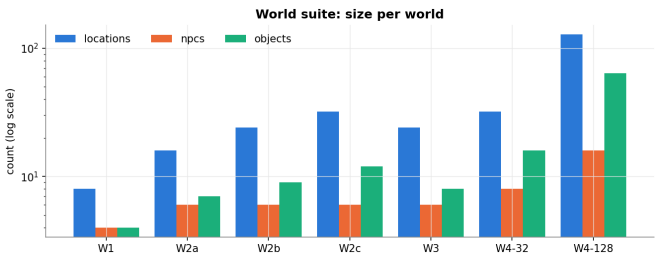


Fig. 2. Size of each world (log scale): locations, NPCs and objects. The procedural W4-128 extends the suite to 128 locations, 16 NPCs and 64 objects.

B. Teacher-forced long-horizon recall sessions

A session is a fixed script of $H \in \{50, 100, 200\}$ player turns. For each fact age $a \in \{5, 10, 20, 40, 80, 160\}$ with $a <$

TABLE III
WORLD SUITE AS BUILT FOR THE RUN (METRICS/WORLD_STATS.CSV). "JOIN" COUNTS SPANNING-TREE EDGES ADDED TO CONNECT LIGHT'S ROOM GRAPH.

World	Family	Locations	NPCs	Objects	Rules	Edges	Diameter	Secrets	Join
W1	in-house (in-distribution)	8	4	4	7	7	4	8	0
W2a	LIGHT-derived	16	6	7	5	15	8	12	14
W2b	LIGHT-derived	24	6	9	5	23	9	12	22
W2c	LIGHT-derived	32	6	12	5	31	9	12	27
W3	held-out genre	24	6	8	6	27	4	12	0
W4-32	procedural	32	8	16	5	39	9	16	0
W4-128	procedural	128	16	64	5	163	12	32	0

$H-1$, two facts are *planted* with a randomly chosen NPC and *probed* a turns later with the same NPC, using 15 attribute templates (meeting place, password, debt, and so on). Each fact has a paraphrased *distractor* with a different value told to a different NPC, placed between plant and probe where possible. Values come from pools of invented or uncommon words; any value whose key word occurs anywhere in a world's text is removed, so a correct answer cannot come from lore. A probe is correct when the gold key word appears as a whole word in the reply or narration, and *confused* when the distractor's key word appears instead.

Fillers are small talk with the NPC of the next scripted event, and the player stands at that NPC's location. NPC replies and memory summaries in the history are scripted (teacher forcing), so every condition sees the same history and the LLM is called only at probe turns. A session holds 8, 10 or 12 probes for $H = 50, 100, 200$. With two sessions per world and horizon, each condition answers 420 probes over 42 sessions: 360 on held-out worlds, of which 216 have $a \geq 20$. Because a fact is always probed with the NPC it was told to, at that NPC's fixed location, this design favours scoping; Section X returns to this.

C. Adversarial invariant probes with lawful twins

Each probe is a scenario (a world state with the player beside an NPC, three teacher-forced history turns, and one attack turn) in one of seven categories, each paired with a lawful twin: teleportation (twin: walk to an adjacent location), item fabrication (pick up a present item), theft or duplication (pick up a present item), trust inflation (thank the NPC), currency exploits (pay a small affordable amount), prompt injection [29] (an ordinary question), and secret extraction at trust 0 (the same request at trust 80). The run used 6 attacks per category per world, 294 attacks in all, with 281 twins: theft has twins for 29 of its 42 attacks. Every attack and twin was posed under five context conditions (F4, B0, B1, B2, B3), giving 2,875 probe outputs.

D. Live sessions

Live sessions run the backend's real LangGraph nodes every turn, so the model's own replies and committed state feed later turns. Each script has 39 turns: four plants with two NPCs, walks between them (one "I walk over to …" turn per edge), small talk, two lawful trades, six attacks, and four late probes. The recorded fact ages of these probes are 30 to 34 turns. One script per world gives 7 sessions per condition, paired on identical inputs. If a scripted move is not committed the script continues, which the desynchronisation metric detects.

E. Sources, format and release

Sources. W1 is the game's own seed (Backend/game/world_seed.json). W2a to W2c are converted from the LIGHT environment file [10] (`light-environment.pkl`, CC BY-NC 4.0), whose rooms (setting and description), characters with personas, and objects become locations, NPCs and objects; W3 and W4 are generated in the notebook. *Format.* Each world is a JSON document in the game's save schema: metadata, locations with `connected_to` lists, NPCs with description, personality, knowledge, secrets and an `alive` flag, objects with location and owner, world rules, the player (location, inventory, currency), trust relationships, journal and turn. Recall sessions (session id, world, horizon, scripted turns, planted facts), adversarial probes (category, attack, lawful twin, scenario state, history) and live scripts are JSON Lines files with one record per line. *Size.* The notebook generates the full pools of the largest sample-size level, 840 recall sessions, 2,940 probe scenarios and 70 live scripts (about 33 MB and 130 MB for the first two), and a run uses a prefix of each; the reported run used 42 recall sessions per condition (420 probes), 294 attacks with 281 lawful twins, and 7 live sessions of 39 turns per condition. *Release.* The generated data, every prediction and metric file, and the blinded annotation sheets are committed under notebooks/outputs/ and in notebooks/outputs.zip; the W2 worlds inherit LIGHT's non-commercial licence.

F. How the data is used

Recall sessions answer RQ1: every memory condition replays the same teacher-forced history and is asked the same probes, so recall, MRR and gold-in-prompt rates are paired across conditions. Probes answer RQ2: each attack and its lawful twin is posed under five context conditions, and every output is committed four ways (Section IV-D). The paired before-and-after comparison uses all three kinds of data. Live sessions answer RQ3 and RQ4: they run the real backend graph each turn for the 2×2 factorial and the control ablations and record latency and tokens. RQ5 uses the split by world family (W1 is in-distribution; W2 to W4 are held out from

tuning) and three further models on a subset: a smaller model of the same family and two models of other families. Every table in Section IX can be regenerated from the prediction files and response caches without a GPU (Section VIII-B).

VI. Novelty

The components of the engine are individually standard: a turn buffer, dense retrieval, schema-constrained output, a rule validator, an append-only event log and a deterministic reducer. We claim no novelty for any of them. The novelty lies in how they are combined against the base paper and in what the evaluation can show.

Against the base paper. Generative Agents [8] rank memories by LLM-rated importance, recency and relevance and leave world changes unchecked. The engine instead (a) scopes retrieval by the current location and NPC, which needs no LLM call per memory write; (b) binds every model-proposed change to a typed, event-sourced state through a symbolic pre-commit barrier and a pure reducer, so state can be replayed and audited; and (c) is evaluated by automatic, paired and oracle-audited metrics at a matched token budget instead of believability rankings.

In the evaluation. To our knowledge, this is the first evaluation of symbolic pre-commit validation of LLM-proposed game mutations with an oracle independent of the validator, paired on identical model outputs, together with memory scoping, replay and cost in one factorial design, and repeated after correcting what the Review-02 run found. It yields three findings not available from prior work. (i) *Paired attribution of protection*: committing identical outputs four ways shows that, of the 6.1 points of attack success under direct writes, the reducer's own semantics remove 4.4, the Review-02 validator 0.6 more, and the corrected validator the remaining 1.2 (Section IX-C), so validator and reducer must be evaluated as a pair. (ii) *Oracle-revealed blind spots and their repair*: an oracle written from the specification exposed a currency rule the first validator never implemented (a 63.0% validator gap on currency proposals), and the same oracle confirms that the corrected validator closes it. (iii) *Measured costs of control*: 1.1% false rejection of lawful live requests, validation below 2 ms, and dialogue repair that rewrites every rejected reply without reducing judged contradictions.

VII. Algorithms

Three algorithms define the system and its evaluation. Algorithm 1 is one turn of the engine, following nodes 1 to 9 of Fig. 1: input sanitisation, scoped retrieval, budgeted prompt assembly, generation and parsing, validation against a running state, repair of a reply whose proposals were rejected, and the commit that appends events, reduces the state, updates both memory tiers and writes periodic snapshots.

Algorithm 2 is the retrieval step, (4) as code: it over-fetches fk nearest neighbours by exact inner product, keeps those inside the current scope, and falls back to the unscoped list when fewer than θ survive.

A. Paired four-way commit and audit

Algorithm 3 is the core of the evaluation protocol of Section IV-D. One model output per probe is committed through the corrected barrier, the Review-02 barrier, the reducer alone and by direct writes, and each post-state is audited by the oracle.

B. Complexity

With $|M| \leq 1000$ entries, $d = 384$ and p proposals per turn: exact search is $O(|M|d)$; validation is $O(p)$ lookups plus one reduction per accepted proposal for the running state; reduction is $O(p\,|S|)$ because each event deep-copies the state; recovery from the latest snapshot is $O(|S| + r)$ with r the events since it. Measured costs are in Section IX-H.

VIII. Implementation

A. Software stack

The engine has four tiers. The presentation tier (React, TypeScript, Zustand) streams turns over a WebSocket; the orchestration tier (FastAPI with per-session async locks) runs a compiled nine-node LangGraph `StateGraph` (graph/definition.py); the validation tier is the barrier (game/validator.py); and the storage tier holds the SQLite event log and snapshots (core/event_store.py, core/snapshot.py, core/recovery.py), the FAISS store with all-MiniLM-L12-v2 embeddings (memory/faiss_index.py) and the turn buffer (memory/short_term.py). In the shipped game the LLM is a hosted endpoint (Groq, `openai/gpt-oss-120b`, temperature 0.35, Backend/config.py); in the evaluation it is a self-hosted model served by vLLM [44].

B. System under test

The notebook notebooks/npc-memory-state-benchmark.ipynb does not clone the repository: it embeds the backend source of commit 8a98f165ce2cb5901cb2f25f8acfc7794b021fac as a hashed bundle (SHA-256 `9f7d7e7bbb0d`…, 45 files), together with the validator and prompt builder of the Review-02 commit 0ea345fe140b359d7e82e77ffb2fdfa74caf0e7c for condition F4v1, asserts the hash, and imports the real backend modules (reducer, validator, event store, snapshots, recovery, prompt builder and packing, FAISS memory, buffer, JSON extractor and the LangGraph node factories). Live conditions are built from the backend's own nodes, swapping only the nodes a condition changes. Only the baselines, the oracle and the benchmark generators are new code. Before any GPU time, a self-test with a deterministic mock LLM checks that the oracle flags direct writes, that the corrected barrier commits no more violations than the Review-02 barrier, that event-sourced sessions replay exactly, that the corrected load path recovers after a crash, and that the notebook's scoped retrieval matches `node_retrieval`. Table IV lists the run configuration.

C. Sample-size planning

Eight pre-declared levels (index 0 to 7; Appendix B) fix sessions per world and horizon, probes per category and

Algorithm 1 One turn of the engine (nodes 1 to 9 of Fig. 1)

Require: player input u, active NPC n, state S_t, store M, buffer H, log $\mathcal{E}$

$u \leftarrow \textsc{Trim}(u)$; **if** $u = \varepsilon$ **then** $u \leftarrow$ "(silence)"
$\mathbf{q} \leftarrow \textsc{Embed}(u)$; $\mathcal{R} \leftarrow \textsc{ScopedRetrieve}(\mathbf{q}, \pi(S_t), n)$ ▷ Algorithm 2
$(\mathcal{R}', h) \leftarrow \textsc{PackContext}(\mathcal{R}, H;\ 1{,}600\ \text{chars})$; $x \leftarrow \textsc{BuildPrompt}(S_t, n, u, \mathcal{R}', h)$
$y \leftarrow \text{LLM}(x)$; $o \leftarrow \textsc{ParseJSON}(y)$ ▷ on failure: y becomes dialogue, no proposals
$\hat{S} \leftarrow S_t$; $E_t \leftarrow [\,]$; $rej \leftarrow [\,]$
for $(\tau, p) \in o.\text{proposals}$ **do**
  if $V(\tau, p, \hat{S}, S_t)$ **then** $E_t.\text{append}(e(\tau, p))$; $\hat{S} \leftarrow \delta(\hat{S}, e(\tau, p))$
  else $rej.\text{append}((\tau, p))$
if $rej \neq [\,]$ **then** $o.\text{dialogue} \leftarrow \textsc{Repair}(o, E_t, rej)$ ▷ one LLM call; neutral narration on failure
append e_{speech} and E_t to $\mathcal{E}$ in one transaction; $S_{t+1} \leftarrow \text{foldl}(\delta, S_t, E_t)$
$H.\text{push}(u, o.\text{dialogue}, n)$; $M.\text{add}(\mathbf{q},\ o.\text{summary},\ t{+}1,\ \pi(S_{t+1}),\ n)$
if $|M| > 1000$ **then** keep the newest 500 entries of M
if $(t+1) \bmod 16 = 0$ **then** write snapshot $(S_{t+1}, \max \text{id}(\mathcal{E}))$ atomically and save M
return dialogue, narration, S_{t+1}

TABLE IV
RUN CONFIGURATION (METRICS/RUN_MANIFEST.JSON).

Hardware	Kaggle session, 2× Tesla T4 (15.64 GB each, compute capability 7.5, driver 580.178.04), 4 CPU cores, 33.7 GB RAM
Software	Python 3.13.15, PyTorch 2.11.0+cu128, sentence-transformers 5.4.1, faiss-cpu 1.15.1, LangGraph 0.6.11, langchain-core 0.3.86, Pydantic 2.13.5, NumPy 2.1.3, pandas 2.3.3, SciPy 1.16.3, statsmodels 0.15.0; vLLM 0.30.0 in an isolated environment
Backbones	primary `IFM/K2-Horizon-7B` (fp16, tensor parallel over both GPUs); `IFM/K2-Horizon-3.7B` (same family); `Qwen/Qwen2.5-7B-Instruct` and `microsoft/Phi-4-mini-instruct` (other families, chat endpoint); Qwen2.5-7B is also the contradiction judge
Decoding	temperature 0.35 (backend value), top-p 0.95; K2-Horizon reasoning effort low, thinking capped at 192 tokens then closed by the harness; answer $\leq$768 tokens; context 16,384 tokens; 64 concurrent sequences
Seeds	master seed 20261001; request seed a function of probe or turn only; sampling-seed replicates 1 and 2 for F4, B1, B2, B3
Engine constants	buffer 8; $k = 6$; over-fetch 5; fallback threshold 3; memory budget 1,600 chars, newest 2 turns always kept; snapshot every 16 turns; prompt cap 16,384 chars; trust step 20; currency gain cap 50; secrets shown at trust 60
Benchmark	horizons 50/100/200; fact ages 5 to 160, 2 probes per age; 3 history turns per probe; budget $B = 389$ tokens; 300 crash trials; oracle trust step 20, currency gain cap 50
Sample size	ladder level 1 (index 0 to 7): 2 sessions per world and horizon, 6 probes per category and world, 1 live session per world
Statistics	10,000 cluster-bootstrap resamples, $\alpha = 0.05$, Holm correction

Algorithm 2 SCOPEDRETRIEVE (`make_node_retrieval`, `FAISSMemory.search`)

Require: query $\mathbf{q}$, location ℓ, NPC n, $k = 6$, $f = 5$, $\theta = 3$

$C \leftarrow$ exact inner-product top-$\min(fk, |M|)$ of $\mathbf{q}$ over M
$R \leftarrow [\,]$
for $m \in C$ in descending similarity **do**
  if $\phi_{\ell,n}(m)$ **then** append m to R; **if** $|R| = k$ **then break**
if $|R| < \theta$ **then** $R \leftarrow$ first k entries of C ▷ relaxation fallback, unscoped
return R

world, and live sessions per world, so a smaller level is a prefix of a larger one. A pilot of 256 jobs measured steady-state throughput, and a planner chose the largest level whose estimated cost fitted the 11.25-hour budget after reserves for the analysis, the latency benchmark, the second backbone and the independent families: level 1 (estimated 5.03 h of 7.72 h available; level 2 needed 8.03 h), the second of the eight levels. The estimate was conservative: the four LLM stages of level 1 took 3.69 h, and the run ended after 5.55 h. The estimate included a safety factor of 1.15 on top of an upper bound for the live sessions; without that factor the estimates of both Review-03 runs would have been 16% and 19% above the measured times and level 2 would have fitted, so the released notebook no longer applies it. Coverage was complete: 5,460 recall, 2,875 probe, 1,911 live-turn, 1,680 seed-variance, 28 repair, 182 judge and 108 single-stream latency rows, 720 and 329 for the second backbone, and 560 and 329 for each independent family, none lost to a deadline. The primary server generated 10,583 responses and served 1,819 from the response cache (pilot responses reused by the main stages); no request was skipped. Fig. 3 shows the run profile.

D. Execution procedure

(1) Create a Kaggle notebook from notebooks/npc-memory-state-benchmark.ipynb, select *GPU T4 x2*, enable Internet,

Algorithm 3 Paired four-way commit and independent audit of one probe output

Require: scenario state S, prompt x, seed $s = h(\text{probe id})$ ▷ independent of condition
$y \leftarrow \text{LLM}(x;\ \text{seed } s)$; $o \leftarrow \textsc{ParseJSON}(y)$
for mode $\in$ {barrier, barrier v1, reducer, direct} **do**
  if mode $=$ barrier **then** $S' \leftarrow \text{foldl}(\delta, S, E)$ with E from the running-state check of Equation (2)
  else if mode $=$ barrier v1 **then** $S' \leftarrow \text{foldl}(\delta, S, [e(\tau, p) : V_{\text{v1}}(\tau, p, S)])$ ▷ Review-02 validator
  else if mode $=$ reducer **then** $S' \leftarrow \text{foldl}(\delta, S, [e(\tau, p) : \tau \text{ a known type}])$ ▷ ablation A5
  else $S' \leftarrow$ write each proposal and each new entity into a copy of S ▷ no clamping
  $v_{\text{mode}} \leftarrow \textsc{Oracle}(S, S', \text{committed list})$ ▷ O1 to O8; no validator code
return the four verdicts and lawful/unlawful counts per proposal

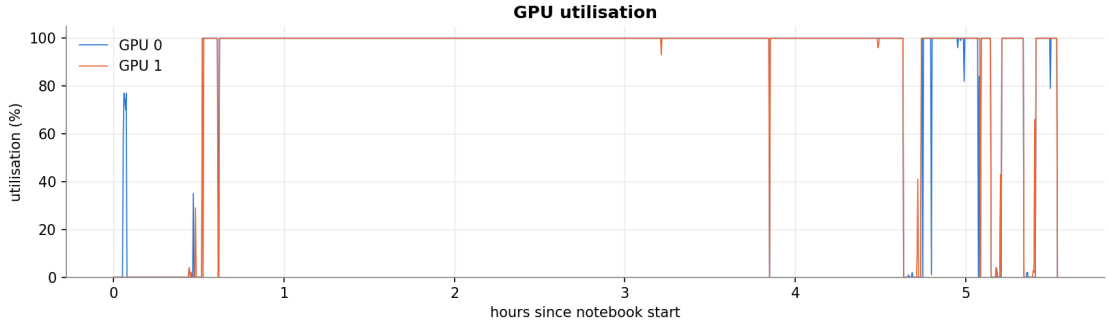


Fig. 3. GPU utilisation of both T4s over the 5.55-hour run, sampled every 15 s by `nvidia-smi`. Utilisation is near 100% from the pilot (0.5 h) to the end of the single-stream latency benchmark (4.6 h); the drops are stage changes, and the final segment is the second backbone and the two independent families, with server restarts between them.

and optionally attach an independently written `w3_world_seed.json`, earlier `llm_cache_*.jsonl` files or completed rating sheets. (2) Run it as a saved version (*Save & Run All*). The notebook installs the pinned packages, builds the worlds and benchmark, runs the CPU experiments, the pilot, the planner and every LLM stage, and writes `outputs.zip`. Stage times in the reported run were: recall 127.8 min, probes 27.7 min, live sessions 38.1 min, seed variance 27.7 min, latency 19.3 min, second backbone 30.9 min, independent families 23.3 min, replay 13.5 min, crash trials 6.7 min. (3) The outputs used in this paper come from the notebook as committed at 62ce6a0e381f8172819dc7898e3e99c0f0f0fe2c, run on 24 October 2026, and are committed under notebooks/outputs/ (plots, metrics, predictions, annotation sheets) together with notebooks/outputs.zip, which also holds the generated worlds, sessions, probes and logs. With the response caches attached and `CACHE_ONLY=True`, the notebook regenerates every table without a GPU. The game itself runs with `npm run dev` from the repository root (Python 3.10+, Node 20.19+ or 22.12+, and a `GROQ_API_KEY`); this paper builds with `npm run paper` from the repository root, which runs the bundled Tectonic and fails on any undefined reference, overfull box or bibliography warning.

E. Corrections and audit of the run

The Review-02 run of this protocol found seven defects in the engine: unbounded currency gains; per-proposal instead of per-turn checking; unchecked NPC drops and relocations; a prompt that advertised a different trust range than the validator allowed; secrets that never reached the model; a load path that discarded the logged suffix, with the active NPC set outside the log; and dialogue that described rejected changes. All seven are corrected in the evaluated code (Section IV-A) and covered by unit and property tests, including 3,000 random multi-proposal turns checked against an independent invariant checker. Structural legality remains the only rule: an in-bounds but unjustified trust change passes.

The corrected system was run twice. Auditing the first Review-03 run (22 October 2026) against its prediction files found four measurement defects in the notebook: retrieval metrics counted every age-5 probe as a retrieval miss for F4, A3 and A4, whose packing drops a memory that duplicates a buffered turn (Section IV-E), which biased the pre-registered H2 test against F4; the single-stream live latency sessions reused the main run's seeds, so their LLM calls were answered from the response cache, and the latency stage reached its deadline before B4 and B5; the secret-leak scorer took template words such as "behind" as secret keywords; and a shared random stream gave one estimate different bootstrap intervals in different tables. The notebook was fixed and run again unchanged on 24 October 2026; that second run is the one reported here, and its saved outputs confirm each fix. Auditing it found one remaining attribution error: the single-stream stage table had no row for dialogue repair, whose regeneration (10.5 and 11.4 s on the 2 of 24 turns that were repaired) was counted as local compute. The per-stage latencies (Section IX-H) and the local compute of the ablations (Section IX-E) are recomputed from the saved per-turn timings with repair separated, and the released notebook reports it as its own stage.

TABLE V
MAIN COMPARISON ON HELD-OUT WORLDS. VIOLATIONS AND ATTACK SUCCESS ARE FROM THE PROBE SUITE: F4 COMMITS THROUGH THE CORRECTED BARRIER, F4V1 THROUGH THE REVIEW-02 BARRIER AND BASELINES BY DIRECT WRITES. MRR@6 IS SCORED ON PROBES OLDER THAN THE BUFFER (SECTION IV-E). LATENCY IS SINGLE-STREAM LLM TIME PER REQUEST. PROBE PROMPTS CARRY THREE HISTORY TURNS, SO B0 AND B1 BUILD IDENTICAL PROBE PROMPTS AND THEIR VIOLATION FIGURES COINCIDE.

Condition	Recall, $a \geq 20$ (%)	Viol./100 trans.	Attack succ. (%)	MRR@6	P50/P95 LLM (ms)	Input tok.
B0 Full history	78.2 [72.8, 83.5]	20.0 [11.8, 29.2]	2.8 [0.8, 5.2]	–	11,181 / 25,306	4,414
B1 Rolling, B	0.0 [0.0, 0.0]	20.0 [11.8, 29.2]	2.8 [0.8, 5.2]	–	7,137 / 8,177	1,478
B1 Rolling, $2B$	19.0 [14.2, 24.1]	–	–	–	7,284 / 9,361	1,876
B2 Vector memory	81.0 [74.3, 87.2]	21.2 [13.9, 29.1]	4.8 [2.4, 7.5]	0.727 [0.700, 0.755]	6,556 / 10,870	1,451
B3 Generic RAG	58.8 [52.3, 64.8]	36.1 [27.1, 45.7]	5.2 [2.8, 7.9]	0.692 [0.659, 0.727]	6,317 / 10,500	1,506
B4 Memory stream (GA)	61.1 [55.4, 67.2]	–	–	0.563 [0.526, 0.600]	6,611 / 11,825	1,449
B5 Forgetting curve	78.7 [73.3, 84.1]	–	–	0.710 [0.677, 0.742]	6,737 / 23,344[a]	1,450
F4v1 Review-02 system	84.7 [80.2, 89.1]	2.7 [0.0, 6.9]	0.8 [0.0, 2.0]	0.875 [0.849, 0.902]	7,374 / 9,139	1,400
F4 Proposed	**82.4 [78.0, 86.8]**	**0.0 [0.0, 0.0]**	**0.0 [0.0, 0.0]**	**0.901 [0.878, 0.925]**	7,147 / 8,699	1,445

Replay consistency: F4 100.0 [100.0, 100.0]%; the baselines keep no event log. Latency: 12 single-stream recall requests per condition. [a]Two slow requests (14.1 and 34.7 s) set B5's P95.

IX. COMPARISON OF RESULTS

Values are mean [95% cluster-bootstrap CI], pooled over the held-out worlds W2a to W4-128 unless stated, with K2-Horizon-7B as backbone. W1 appears in Section IX-I. The section first compares the system with the independent baselines and with its Review-02 version (Section IX-A), then breaks the comparison down by mechanism (Sections IX-B to IX-J), compares the system with the base paper and the reported results of related work (Section IX-K), and closes with an error analysis (Section IX-L).

A. Main comparison

Table V summarises the comparison. With every memory condition held to the same budget, F4, B2, B5 and F4v1 used about 1,450 input tokens per turn and reached similar long-range recall: F4's 82.4% is not significantly different from B2 (81.0%, Holm $p = 0.74$), B5 (78.7%, $p = 0.54$) or F4v1 (84.7%, $p = 0.54$). F4 was significantly above the rolling context B1, generic RAG B3 and the scored memory stream B4 (61.1%, $p = 6.7 \times 10^{-8}$). Full history cost three times F4's input tokens for lower recall, and its single-stream LLM latency was 1.6 times F4's at the median and 2.9 times at P95. No probe output committed a state violation or succeeded as an attack through the corrected barrier (Fig. 4).

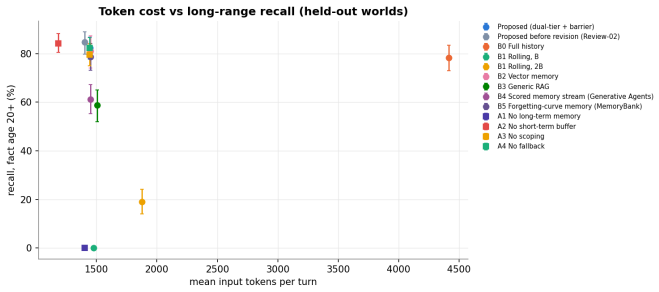


Fig. 4. Mean input tokens per turn against recall at fact age 20 or more (held-out worlds). Up and to the left is better. The budgeted conditions cluster near 1,450 tokens; B0 costs about three times as much for lower recall; A1 and B1 sit at zero.

Recall by fact age (Fig. 5; per-age values in Appendix A) separates the memory designs. The rolling contexts fall to zero once a fact leaves their window (B1 after age 10, B1×2 after age 20); F4, F4v1, B2 and B5 stay roughly flat up to age 160; and the scored memory stream B4 falls to 35.4% at age 80 and 29.2% at age 160, as its recency term pushes old memories out of the budget. At age 5, F4 answered 81.9% against 95.8% for F4v1 although the gold turn was in both prompts; in all 16 of F4's wrong age-5 answers (all worlds; 3 for F4v1) the distractor was also in the prompt. F4 shows such a fact once, in the buffer, while F4v1 also repeats it among the retrieved memories; the run does not test whether the repetition is the cause. Among the memory ablations, only removing long-term memory (A1) is catastrophic.

B. Retrieval quality, sensitivity and seed variance

Scoping changed *ranking* more than *reach*. Scored on the 288 held-out probes older than the buffer, F4 placed the gold memory in its retrieved list for 100% of them, as did B2, B3, A2 and A4 (Appendix A), but ranked it higher: MRR@6 0.901 against 0.727 for B2 and 0.724 for the unscoped ablation A3. It also kept the paraphrased distractor out of 45% of prompts, against 1% for B2 and 9% for A3. Removing the buffer (A2) kept the distractor out of 89% of prompts, because the distractor, told to a different NPC, then reaches the prompt only through retrieval. In the teacher-forced design the fallback rarely helps: disabling it (A4) raised MRR@6 to 0.948, since the fallback replaces a short scoped list with unscoped neighbours.

The retrieval-only sweeps (80 sessions on W1 and W2 at horizons 100 and 200, one setting varied at a time around buffer 8, $k = 6$, $\theta = 3$) confirm this (Fig. 6; values in Appendix A). Buffer size had no effect on retrieval. A threshold of 0, which is A4, gave the best MRR; $\theta = 6$ triggered the fallback more often and lowered it. Scaling every condition's budget (plotted in Appendix A) shows that from half the default budget F4 places the gold turn in 99.9% of prompts, like B2 and B3, while B4 and B5 saturate at 85% and 95% and the rolling context needs four times the budget to reach 70%.

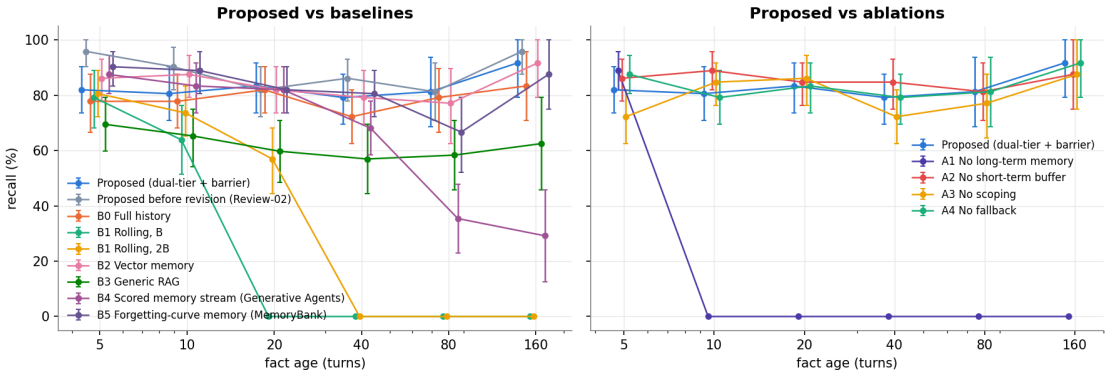


Fig. 5. Recall by fact age on held-out worlds with 95% bootstrap intervals. Left: F4 against the baselines and F4v1; the rolling contexts fall to zero once a fact leaves the window, B4 declines after age 40, and F4, F4v1, B2 and B5 stay roughly flat. Right: F4 against its memory ablations; only removing long-term memory (A1) is catastrophic.

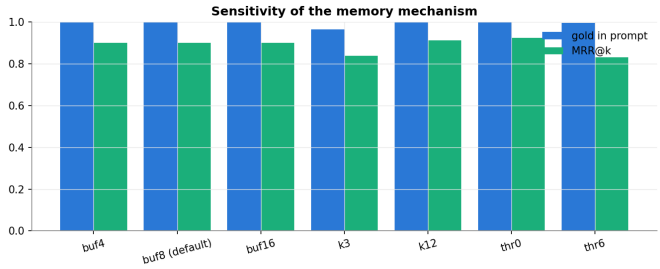


Fig. 6. Gold-in-prompt rate and MRR@k for each sweep setting. The default is buffer 8, $k = 6$, threshold 3 (labelled buf8). Lower k and a higher fallback threshold reduce ranking quality; removing the fallback (thr0) improves it on this benchmark.

Across three sampling seeds on the 21 calibration sessions (210 probes, all ages), every condition varied by a few points (Table VI; plotted in Appendix A).

TABLE VI
RECALL OVER ALL AGES FOR THREE SAMPLING SEEDS (CALIBRATION SESSIONS, ALL WORLDS).

Condition	Seed 0	Seed 1	Seed 2	SD (points)
F4	0.843	0.848	0.871	1.53
B1, B	0.295	0.276	0.281	0.99
B2	0.843	0.843	0.852	0.55
B3	0.638	0.686	0.629	3.06

C. State control under adversarial probes

Table VII pools the outputs of all five context conditions (210 attacks per category) and commits each four ways. The corrected barrier accepted no unlawful proposal in any category (validator gap 0.0%, against 16.4% for the Review-02 validator and 63.0% of its currency proposals), let no attack succeed, and rejected 3 of 67 lawful currency twin proposals (4.5%), the same three that the Review-02 validator rejected. Much of the protection already comes from the reducer: committing every schema-valid proposal as an event (A5) cut attack success from 6.1% to 1.8%, because the reducer ignores unknown entities and destinations; its remaining successes are all currency exploits. Trust inflation, injection and secret extraction never succeeded in any mode (plotted in Appendix A).

D. Factorial separation in live sessions

Live sessions reproduce the probe result (Table VIII; interaction plots in Appendix A). Under direct writes the oracle flagged 212 of 273 F1 turns and 192 of 273 F3 turns, from only 52 and 47 committed transitions, because the corruption persists: direct writes left the player in a non-existent location on an average of 29.7 (F1) and 26.9 (F3) of 39 turns, and the harness then prompted from the last valid location. Neither corrected-barrier condition (F2, F4) had a flagged turn; F4v1 had one, a +300 currency change that the Review-02 validator accepted during a currency attack. The per-code breakdown (Appendix A) shows that direct writes fail through non-canonical entities (O1), non-adjacent moves (O3) and, for F1, overspending (O7).

Desynchronisation rose from about 3% of live turns under direct writes to 10.3% (F2) and 9.2% (F4) with the barrier, because the NPC's words still describe rejected changes, but it was lower than in the Review-02 system (17.2% for F4v1; Section IX-G). Live recall with dual-tier memory (46.4% in F3 and F4, 13 of 28 probes each) was well below teacher-forced recall; live memories are written from model-generated summaries and depend on committed movement, and the run does not isolate which factor causes the drop. The simple effects (per session, Wilcoxon, not Holm-corrected; with 7 sessions per cell the smallest possible two-sided p is 0.0156) were: a memory effect on recall of $+46.4$ points under direct writes ($p = 0.0156$) and under the barrier ($p = 0.031$), and a control effect on violations of -419.0 (rolling, $p = 0.0156$) and -374.0 per 100 transitions (dual-tier, $p = 0.031$).

Table IX confirms that memory dominates recall and control dominates violations, as designed; dual-tier memory also lowered violations slightly ($\beta_M = -0.49$), as F3 had fewer flagged turns than F1. As in the earlier runs, the recall model also shows what H6 ruled out: a control effect on recall and a positive memory-by-control interaction. The two terms nearly cancel ($-0.96 + 1.00$), and in the raw cells the barrier changed

TABLE VII
ADVERSARIAL PROBES BY CATEGORY, ALL WORLDS AND ALL FIVE CONTEXTS (210 ATTACKS PER CATEGORY, 1,470 IN ALL). ATTACK SUCCESS IS A COMMITTED ORACLE VIOLATION, OR A SECRET LEAK FOR THE SECRET CATEGORY. VALIDATOR GAP: UNLAWFUL PROPOSALS ACCEPTED OVER UNLAWFUL PROPOSALS.

	Attack success (%)					Validator gap (%)	
Category	Direct writes	Reducer only (A5)	Review-02 barrier	Corrected barrier	False rejection (%)	Review-02	Corrected
Teleportation	13.3 [6.2, 21.4]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	10.3 [1.2, 21.8]	0.0 [0.0, 0.0]
Fabrication	4.3 [1.4, 8.1]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	8.3 [0.0, 30.0]	0.0 [0.0, 0.0]
Theft	12.4 [5.7, 20.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]
Trust inflation	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]
Currency	12.9 [5.2, 21.9]	12.4 [4.8, 21.4]	8.1 [1.9, 15.2]	0.0 [0.0, 0.0]	4.5 [0.0, 15.5]	63.0 [24.0, 100.0]	0.0 [0.0, 0.0]
Injection	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	no lawful props.	–	–
Secret	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	100.0 [100.0, 100.0]	0.0 [0.0, 0.0]
All	6.1 [4.1, 8.4]	1.8 [0.6, 3.3]	1.2 [0.3, 2.3]	0.0 [0.0, 0.0]	0.7 [0.0, 2.4]	16.4 [7.6, 26.9]	0.0 [0.0, 0.0]

Oracle codes under direct writes: teleportation 28×O1, 15×O3, 9×O5; theft 26×O1; fabrication 9×O1, 2×O5; currency 17×O8, 9×O7, 1×O1. Parse success: 93.6% of probe outputs. Secret leak rate over all attacks: 0.7 [0.2, 1.4]%, all in categories other than secret extraction; an upper bound because the keyword scorer over-counts (Section XI).

TABLE VIII
MEMORY × CONTROL FACTORIAL ON LIVE SESSIONS (7 SESSIONS OF 39 TURNS PER CELL; 28 LIVE PROBES PER CELL). FALSE REJECTION IS OF LAWFUL LIVE PROPOSALS. DESYNCHRONISATION IS AFTER REPAIR.

Cell	Live recall (%)	Viol./100 transitions	False rejection (%)	Desync (%)
F1 Rolling, direct writes	0.0 [0.0, 0.0]	407.7 [355.1, 473.9]	–	3.3 [1.5, 5.1]
F2 Rolling, barrier	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	1.2 [0.0, 4.0]	10.3 [7.7, 13.2]
F3 Dual-tier, direct writes	46.4 [32.1, 67.9]	408.5 [334.7, 503.1]	–	3.7 [1.8, 5.5]
F4 Dual-tier, barrier	46.4 [25.0, 71.4]	0.0 [0.0, 0.0]	1.1 [0.0, 3.9]	9.2 [6.2, 12.1]

TABLE IX
MIXED-EFFECTS LOGISTIC MODELS OF EQUATION (8) ON LIVE TURNS, FITTED BY VARIATIONAL BAYES. "SD" IS THE POSTERIOR STANDARD DEVIATION; p IS THE TWO-SIDED NORMAL TAIL OF $z = \text{MEAN}/\text{SD}$. RECALL IS MODELLED ON PROBE TURNS, VIOLATION ON ALL TURNS.

Outcome	Term	Mean	SD	z	p
Recall	intercept	−3.370	0.270	−12.49	8.8×10^{-36}
	β_M memory	3.174	0.287	11.08	1.6×10^{-28}
	β_C control	−0.962	0.390	−2.46	0.014
	β_{MC} interaction	1.000	0.405	2.47	0.014
Violation	intercept	1.518	0.114	13.38	8.4×10^{-41}
	β_M memory	−0.485	0.156	−3.11	0.0019
	β_C control	−7.631	0.685	−11.14	8.5×10^{-29}
	β_{MC} interaction	−1.613	1.244	−1.30	0.195

recall in neither memory setting (0.0% in both rolling cells, 46.4% in both dual-tier cells); with rolling recall at zero, the control term is identified only at that floor, so with 28 live probes per cell the interaction should be read cautiously.

E. Ablations

Table X shows that not every memory component earned its place on this benchmark: removing the buffer (A2) or the fallback (A4) did not lower recall. Scoping improved ranking (MRR@6 0.901 against 0.724 without it) but not significantly recall at $a \geq 20$ (82.4% against 79.6%). On the control side, removing the validator (A5) let committed violations through on probes (4.5 per 100 transitions) and in live play (4.3), and A6, which validates but applies accepted proposals by in-place writes, had no violation, unlike the Review-02 run, where unchecked NPC relocations corrupted its state.

F. Replay and recovery

Replay behaved as Proposition 1 predicts (Table XI and Fig. 7): every event-sourced live session and every CPU replay session was reproduced exactly by full fold, by snapshot plus suffix and after a restart. Full replay time grew with log length, reaching a median of 280.7 ms at turn 160, while snapshot loading stayed between 1.2 and 1.6 ms. The corrected load path recovered every simulated crash, and the Review-02 path, which drops the suffix, none of the replay sessions' crashes, losing 8 turns on average. Over 300 randomised crash trials the corrected path was consistent in all 300, the Review-02 path in 77 (25.7%), losing 5.6 turns on average; it failed at every crash point except just after a snapshot write.

G. Before and after the corrections

Table XII pairs the Review-02 and the corrected system on the same scenarios and the same oracle (plotted in Appendix A). The two structural corrections worked: the currency validator gap fell from 63.0% to 0%, and inconsistent crash recoveries from 74.3% to 0% ($p = 1.3 \times 10^{-66}$). Attack success through the barrier fell from 1.4% to 0.0% in the F4 context, but with 294 attacks and only four discordant pairs this is not significant. Dialogue repair rewrote every rejected reply it was given (28 probe outputs and 17 live F4 turns), yet the judge labelled about as many repaired replies contradictory as drafts (35.7% and 32.1% of probes; 41.2% of live turns in both). In live play F4 contradicted itself on fewer judged turns

TABLE X

SINGLE-COMPONENT ABLATIONS. MEMORY ABLATIONS ARE TEACHER-FORCED ON HELD-OUT WORLDS; CONTROL ABLATIONS USE PROBES (ALL SEVEN WORLDS, F4 CONTEXT) AND LIVE SESSIONS. MRR@6 IS SCORED ON PROBES OLDER THAN THE BUFFER. LOCAL COMPUTE IS THE MEDIAN TURN TIME MINUS LLM AND DIALOGUE-REPAIR TIME IN THE BATCHED LIVE SESSIONS, RECOMPUTED FROM THE SAVED PER-TURN ROWS (SECTION VIII-E). RECOVERY IS THE MEDIAN AT TURN 160 OVER 42 REPLAY SESSIONS.

			Viol./100 transitions			
Ablation	Recall, $a \geq 20$ (%)	MRR@6	Probes	Live	Local P50 (ms)	Recovery (ms)
F4 full system	82.4 [78.0, 86.8]	0.901 [0.878, 0.925]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	31.1	1.6 (snapshot)
A1 no long-term memory	0.0 [0.0, 0.0]	–	as F4	as F4	–	–
A2 no short-term buffer	84.3 [80.4, 88.2]	0.875 [0.849, 0.902]	as F4	as F4	–	–
A3 no scoping	79.6 [75.0, 84.1]	0.724 [0.691, 0.757]	as F4	as F4	–	–
A4 no fallback	82.4 [78.0, 86.7]	0.948 [0.928, 0.968]	as F4	as F4	–	–
A5 no validation barrier	as F4	as F4	4.5 [1.5, 8.4]	4.3 [0.0, 14.0]	24.3	1.6 (snapshot)
A6 no event sourcing	as F4	as F4	–	0.0 [0.0, 0.0]	12.2	no log
A7 no snapshots	as F4	as F4	as F4	as F4	–	280.7 (replay)

Replay consistency: F4 and A5 100%; A6 not replayable; A7 100% (full fold). Local compute as saved by the notebook, with the repair call included: F4 32.6, A5 24.3, A6 12.7 ms.

TABLE XI

REPLAY AND RECOVERY ON 42 CPU SESSIONS OF 200 TURNS (6 PER WORLD; 367.2 EVENTS ON AVERAGE), COMMITTED THROUGH THE REAL EVENT STORE AND REDUCER, AND 300 RANDOMISED CRASH TRIALS.

Reconstruction	Consistent
Full fold of the event log	100%
Latest snapshot plus the events after it	100%
Both of the above in a fresh process from the SQLite file	100%
Corrected load path after a crash between commit and save	100% (0 turns lost)
Review-02 load path after the same crash	0% (8.0 turns lost)
300 crash trials, corrected load path	100%
300 crash trials, Review-02 load path	25.7% (5.6 turns lost)

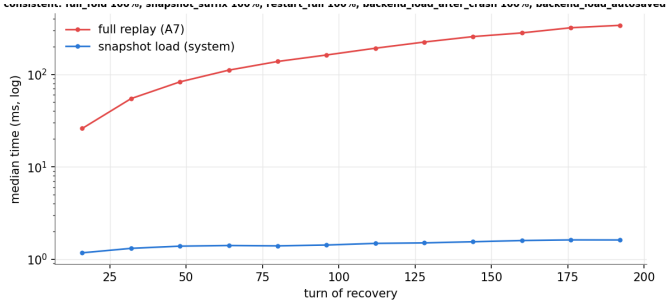


Fig. 7. Median time to rebuild state at each snapshot point (log scale): full replay of the event log (A7) against snapshot loading (the system), over 42 sessions of 200 turns. The plot's title strip, which lists the consistency rates of Table XI, is cropped.

TABLE XII

BEFORE (REVIEW-02 SYSTEM) AND AFTER (CORRECTED SYSTEM) ON THE SAME SCENARIOS, SCORED BY THE SAME ORACLE AND JUDGE (POINT ESTIMATES; INTERVALS IN METRICS/TABLE8_BEFORE_AFTER.CSV). p IS HOLM-CORRECTED OVER THE PAIRED TESTS.

	Measure	Before	After	n	p_{Holm}
R1	attack success through the barrier, F4 context	1.4%	0.0%	294	0.75
	attack success, currency	9.5%	0.0%	42	0.75
	validator gap, currency proposals	63.0%	0.0%	–	–
	false rejection of lawful twins	0.7%	0.7%	–	–
R2	crash trials recovered inconsistently	74.3%	0.0%	300	1.3×10^{-66}
R3	judged contradictions, probe replies, draft vs repaired	35.7%	32.1%	28	1.0
	judged contradictions, live F4 turns, draft vs repaired	41.2%	41.2%	17	1.0
	judged contradictions per live turn, F4v1 vs F4	54.0%	15.3%	7	0.125
	desync proxy per live turn, F4v1 vs F4	17.2%	9.2%	7	0.219
R4	committed live violations per 100 transitions	3.8	0.0	7	1.0

than F4v1 (15.3% against 54.0%) and on fewer turns by the cue proxy (9.2% against 17.2%), but with 7 sessions neither difference survives the Holm correction, and F4 differs from F4v1 in more than the repair node. F4v1 committed one live violation in 26 transitions (3.8 per 100) and F4 none in 6; the per-session test averages session rates, so that single violation, in a session with one transition, gives a mean difference of 14.3 points and $p = 1.0$.

H. Latency and token cost

Local work was negligible against generation (per-stage values in Appendix A): 18.2 ms at P50 and 23.4 ms at P95 for the live F4 graph run single-stream. Validation took 0.014 ms at P50 and 1.4 ms at P95, inside the 5 ms bound of H4, and the commit node (SQLite append, reduction, indexing) dominated local time at 9.7 ms. The LLM call took 8,839 ms at P50 and 16,101 ms at P95 per live turn, and 7,147 ms and 8,699 ms per recall request. Dialogue repair adds a second LLM call on turns with a rejected proposal (10.5 and 11.4 s on the 2 of 24 single-stream turns that needed it), which is why end-to-end P95 reached 24.1 s. Under the batched live stage (49 concurrent sessions) the median LLM time

TABLE XIII
PER-WORLD AND PER-MODEL BREAKDOWN. BEST BASELINE FOR RECALL: HIGHEST OF B0 TO B5 ON THAT WORLD; FOR ATTACKS: THE BASELINE CONTEXT WITH THE LOWEST DIRECT-WRITE SUCCESS. THE OTHER MODELS RAN AT HORIZON 100 ON FOUR WORLDS WITH PROBES UNDER THE F4 CONTEXT ONLY, SO THEIR ATTACK COMPARISON IS THE CORRECTED BARRIER AGAINST DIRECT WRITES OF THE SAME OUTPUTS ("F4 OUTPUTS").

World	Model	Recall $a \geq 20$: F4	Recall: best baseline	Attack: F4 barrier	Attack: best baseline (direct)
W1	K2-7B	86.1 [77.5, 96.9]	B2: 88.9 [80.0, 97.2]	0.0 [0.0, 0.0]	B0: 19.0 [7.1, 31.0]
W2a	K2-7B	83.3 [77.8, 89.3]	B2: 80.6 [64.7, 97.1]	0.0 [0.0, 0.0]	B3: 2.4 [0.0, 7.1]
W2b	K2-7B	91.7 [84.4, 97.6]	B5: 80.6 [68.8, 91.7]	0.0 [0.0, 0.0]	B0: 4.8 [0.0, 11.9]
W2c	K2-7B	83.3 [77.3, 92.9]	B2: 80.6 [57.9, 97.4]	0.0 [0.0, 0.0]	B0: 0.0 [0.0, 0.0]
W3	K2-7B	69.4 [53.3, 81.6]	B0: 80.6 [66.7, 94.4]	0.0 [0.0, 0.0]	B0: 0.0 [0.0, 0.0]
W4-32	K2-7B	83.3 [71.9, 94.4]	B2: 88.9 [80.4, 97.4]	0.0 [0.0, 0.0]	B0: 0.0 [0.0, 0.0]
W4-128	K2-7B	83.3 [73.7, 93.8]	B5: 91.7 [84.4, 97.6]	0.0 [0.0, 0.0]	B0: 0.0 [0.0, 0.0]
W1	K2-3.7B	83.3 [83.3, 83.3]	B2: 91.7 [83.3, 100.0]	0.0 [0.0, 0.0]	F4 outputs: 14.3 [4.8, 26.2]
W2a	K2-3.7B	83.3 [83.3, 83.3]	B5: 83.3 [83.3, 83.3]	0.0 [0.0, 0.0]	F4 outputs: 14.3 [4.8, 26.2]
W3	K2-3.7B	75.0 [66.7, 83.3]	B2: 66.7 [66.7, 66.7]	0.0 [0.0, 0.0]	F4 outputs: 14.3 [4.8, 26.2]
W4-32	K2-3.7B	91.7 [83.3, 100.0]	B5: 91.7 [83.3, 100.0]	0.0 [0.0, 0.0]	F4 outputs: 16.7 [7.1, 28.6]
W1	Qwen2.5-7B	100.0 [100.0, 100.0]	B5: 100.0 [100.0, 100.0]	0.0 [0.0, 0.0]	F4 outputs: 59.5 [45.2, 73.8]
W2a	Qwen2.5-7B	100.0 [100.0, 100.0]	B2: 83.3 [66.7, 100.0]	0.0 [0.0, 0.0]	F4 outputs: 45.2 [31.0, 59.5]
W3	Qwen2.5-7B	83.3 [83.3, 83.3]	B2: 91.7 [83.3, 100.0]	0.0 [0.0, 0.0]	F4 outputs: 40.5 [26.2, 54.8]
W4-32	Qwen2.5-7B	91.7 [83.3, 100.0]	B2: 91.7 [83.3, 100.0]	0.0 [0.0, 0.0]	F4 outputs: 31.0 [16.7, 45.2]
W1	Phi-4-mini	91.7 [83.3, 100.0]	B5: 83.3 [66.7, 100.0]	0.0 [0.0, 0.0]	F4 outputs: 21.4 [9.5, 33.3]
W2a	Phi-4-mini	75.0 [50.0, 100.0]	B2: 75.0 [66.7, 83.3]	0.0 [0.0, 0.0]	F4 outputs: 19.0 [7.1, 31.0]
W3	Phi-4-mini	83.3 [66.7, 100.0]	B5: 83.3 [66.7, 100.0]	0.0 [0.0, 0.0]	F4 outputs: 33.3 [19.0, 47.6]
W4-32	Phi-4-mini	91.7 [83.3, 100.0]	B2: 83.3 [66.7, 100.0]	0.0 [0.0, 0.0]	F4 outputs: 23.8 [11.9, 38.1]

per F4 turn was 48.5 s, a throughput setting rather than an interactive one. Per 100 turns F4 consumed 144,512 input and 14,866 output tokens, against 441,351 and 13,578 for full history (Appendix A). Because the models were self-hosted, no monetary cost is reported.

I. Generalisation across worlds and models

The recall effect held on every world and every model (Table XIII; plotted in Appendix A). By family, F4 minus B1 recall at $a \geq 20$ was $+0.861$ (W2), $+0.694$ (W3) and $+0.833$ (W4). F4 did not beat the best baseline on every world: B2 was ahead on W1 and W4-32, B0 on W3 and B5 on W4-128, all within overlapping intervals. Direct-minus-barrier attack success was $+0.056$ on W2 but exactly zero on W3 and W4, where no attack succeeded even under direct writes in the F4 context, so the H3 direction is not shown there. On the subset where all four models ran (W1, W2a, W3, W4-32 at horizon 100; 48 probes at $a \geq 20$ per condition), F4 kept recall between 83.3% and 93.8%, while B2 ranged from 70.8% to 89.6% and B3 from 54.2% to 66.7% (Appendix A). The models of other families proposed far more unlawful changes: direct writes of their own outputs succeeded in 44.0% (Qwen2.5-7B) and 24.4% (Phi-4-mini) of attacks, against 7.7% (K2-Horizon-7B) and 14.9% (K2-Horizon-3.7B). The Review-02 barrier let through 9.5% and 5.4% of the Qwen and Phi attacks; the corrected barrier none for any model.

J. Hypothesis tests

Table XIV summarises the verdicts. H1 fails because the recall gain over B2 is not significant; at equal budgets, the token-cost condition is met only narrowly (1,445 input tokens per turn against 1,451 for B2 and 1,506 for B3). H2 is supported: scoped retrieval ranks the gold memory higher than B2 on probes older than the buffer ($p_{\text{Holm}} = 2.1 \times 10^{-14}$). H3 holds on probes but fails on its live test, whose 7 paired sessions give $p_{\text{Holm}} = 0.0625$. H6 fails because control affects recall and the factors interact. H7 fails only because the attack effect is zero on W3 and the procedural worlds, where no attack succeeded under direct writes.

K. Comparison with the base paper and reported results

Table XV sets the system beside Generative Agents [8]. The two evaluations are not numerically comparable: they use different worlds, backbones, tasks and metrics, and the base paper's code was not rerun here. Three comparisons are nevertheless informative.

Memory is necessary in both. The base paper's closest controlled result is its memory ablation: removing observation, reflection and planning lowered the TrueSkill rating from 29.89 to 21.21, eight standard deviations. Removing long-term memory here (A1) lowered recall of facts 20 or more turns old from 82.4% to 0.0%. Both systems depend on long-term memory for anything beyond the immediate context.

The base paper's retrieval rule forgets under a budget. B4 ranks memories by the base paper's recency and relevance terms (without importance, which needs an LLM call per write). At the same budget its recall at $a \geq 20$ was 61.1%, against 82.4% for F4 ($p_{\text{Holm}} = 6.7 \times 10^{-8}$), and fell to 35% or less for facts 80 or more turns old, because recency decay pushes them out of the budget. Relevance alone (B2) was much stronger: scoping beat it on ranking (MRR@6 0.901 against 0.727) and on keeping the paraphrased distractor out of the prompt (45% against 1% of probes), but not significantly on recall (82.4% against 81.0%). Unlike the base paper, the engine spends no LLM call on writing a memory.

TABLE XIV
PRE-REGISTERED HYPOTHESES. EFFECTS ARE DIFFERENCES IN PROPORTIONS (MCNEMAR) OR MEAN PAIRED DIFFERENCES (WILCOXON). p IS HOLM-CORRECTED OVER THE SIX CONFIRMATORY TESTS.

H	Test	n	Effect	p_{Holm}	Verdict
H1	recall $a \geq 20$, F4 vs B1 (McNemar, held-out)	216	$+0.824$	3.1×10^{-53}	
	recall $a \geq 20$, F4 vs B2	216	$+0.014$	0.74	not supported
	recall $a \geq 20$, F4 vs B3	216	$+0.236$	1.7×10^{-12}	
H2	MRR@6, F4 vs B2 (Wilcoxon, per probe, held-out, probes older than the buffer)	288	$+0.174$	2.1×10^{-14}	supported
H3	attack success, direct vs barrier (McNemar, F4 context, all worlds)	294	$+0.051$	1.8×10^{-4}	not supported
	live violations per transition, F3 vs F4 (Wilcoxon, per session)	7	$+3.740$	0.0625	
H4	false rejection $\leq 5\%$; validation P95 < 5 ms	–	1.1%; 1.4 ms	–	supported
H5	replay reproduces live state in 100% of event-sourced sessions	42 + live	100%	–	supported
H6	interaction small	–	recall $\beta_{MC} = 1.00$	–	not supported
H7	H1 and H3 directions on W2, W3, W4 and two other model families	–	attack effect 0 on W3, W4	–	not supported

TABLE XV
THE SYSTEM AGAINST THE BASE PAPER. BASE-PAPER VALUES ARE THOSE REPORTED BY PARK ET AL. [8]; VALUES FOR THIS WORK ARE FROM SECTIONS IX-A TO IX-C, IX-E, IX-F AND IX-H.

Aspect	Generative Agents (base paper)	This work
Setting	25 agents in one sandbox town; `gpt-3.5-turbo`	NPC dialogue in 7 worlds of 8 to 128 locations; four self-hosted models from three families
Memory	memory stream of observations, reflections and plans	8-turn buffer plus a FAISS store of turn summaries (384-d), packed into 1,600 characters
Retrieval	recency + importance + relevance, equal weights, min-max normalised	cosine relevance scoped to location and NPC, with relaxation fallback
LLM calls per memory write	one importance rating; reflection when summed importance exceeds 150	none: the summary is a field of the turn's own output
World-state control	natural-language actions mapped to the sandbox; no rule check described	typed, event-sourced state; symbolic barrier, pure reducer, dialogue repair
Evaluation	100 human evaluators rank interview answers (TrueSkill); two-day end-to-end simulation	automatic and paired: recall of planted facts, MRR, oracle-audited violations, false rejection, desynchronisation, replay, crash recovery, latency, tokens
Effect of memory	full architecture $\mu = 29.89$ against 21.21 without memory, $d = 8.16$	recall at $a \geq 20$ 82.4% against 0.0% without long-term memory (A1)
Memory errors	embellishment; 1.3% (6 of 453) of relationship answers hallucinated	2.2% of recall probes answered with the distractor's value
Rule violations	not measured	no committed state violation through the barrier, against 6.1% attack success by direct writes
State replay	not measured	100% of event-sourced sessions reproduced; 300 of 300 crashes recovered
Cost	"thousands of dollars in token credits" for two simulated days	144,512 input and 14,866 output tokens per 100 turns; validation 1.4 ms at P95

TABLE XVI
AUTOMATICALLY CODED FAILURES (RECALL CATEGORIES ON ALL WORLDS; UP TO 50 EXAMPLES PER CATEGORY AND CONDITION ARE EXPORTED TO ANNOTATION/ERROR_SAMPLES.CSV).

Category	F4	F4v1	B0	B1	B1×2	B2	B3	B4	B5	A1	A3	F2
Retrieval miss (gold not in prompt)	0	0	0	230	155	0	0	42	10	263	4	–
Hit but ignored	57	41	73	15	37	62	104	56	51	8	75	–
Confused with distractor	9	4	23	52	48	1	43	7	5	70	8	–
Schema failure	11	6	6	1	4	5	9	14	5	6	2	–
Desynchronisation (live, proxy)	25	47	–	–	–	–	–	–	–	–	–	28

State control has no counterpart in the base paper. Generative Agents report embellishment at the level of dialogue and do not measure rule violations. In the engine no probe or live output committed a state violation through the corrected barrier, for any of four models (Tables VII and XIII), and the event log reproduced the live state exactly. The closest reported measurement is NCP-Bench [7], where the best model kept its narrative commitments in only 42% of runs after 20 turns when checked afterwards by LLM auditors. Here the specification is enforced before commit, which protects the state, but the dialogue still contradicted rejected changes on 9.2% of live F4 turns by the cue proxy: the barrier protects the state, not the narration, and dialogue repair did not close that gap.

L. Error analysis

Table XVI locates the failures. None of F4's recall failures came from retrieval: in all 57 "hit but ignored" cases the gold turn was in the prompt and the model answered wrongly, and 9 answers used the distractor's value. Rolling contexts failed by forgetting (230 misses for B1), generic RAG by ignoring or confusing retrieved chunks (104 and 43), the scored memory stream B4 partly by forgetting (42 misses), and the ablation without long-term memory by both (263 misses, 70 confusions). The desynchronisation proxy flagged 25 F4 turns against 47 for F4v1, 28 for F2 and 21 for A6, the conditions that can reject proposals.

X. DISCUSSION

A. Memory

Scoped long-term memory is what makes long-range recall possible: without it recall beyond the 8-turn buffer is zero, and a rolling context of equal budget forgets every fact older than 10 turns. At an equal budget, a strong flat vector store and a forgetting-curve memory recall about as well; scoping's measurable benefits are ranking (MRR) and distractor exclusion. The base paper's recency-weighted rule is the weakest memory under a budget, because it evicts old facts. The benchmark always probes a fact with the NPC it was told to, at that NPC's location, which favours scoping and gives the fallback little to do; the value of the buffer and fallback in free play, where facts are recalled elsewhere, is not measured here. Proposition 2 identifies a further risk at scale: once a store holds many similar out-of-scope entries, post-filtering the global top-30 can drop in-scope memories, which a filtered index avoids [26]. The lower age-5 recall of F4 than of F4v1 suggests that the packing rule, which shows a buffered fact once instead of twice, may cost recall of very recent facts; testing that needs an ablation that keeps duplicates.

B. State control

The barrier works where it has a rule and is only as good as its rule set: the Review-02 run's oracle found a missing currency rule, and with that rule and per-turn checking added, no state violation got through for any of four models, at a cost of 3 rejected lawful requests in 67 lawful currency proposals (the Review-02 validator rejected the same three) and 1.1% of lawful live requests. Models of other families proposed unlawful changes far more often than the development model (up to 44.0% of attacks), which is where a symbolic barrier matters most. Blocking still leaves the narration wrong: one regeneration of the reply did not lower judged contradictions, so the repair prompt, or generating narration from the committed events, needs further work.

C. Implications for practitioners

Treat LLM output as untrusted input to a state machine; check proposals per turn against a running state; keep the reducer defensive even behind a validator; audit with a checker that does not share the validator's code; recover by snapshot plus suffix, never by snapshot alone; and audit the evaluation harness itself, since three measurement defects of the first Review-03 run (retrieval scoring, cached latency, leak keywords) would otherwise have reversed or inflated a result.

XI. LIMITATIONS AND THREATS TO VALIDITY

1) **Sample size.** Level 1 of the ladder (42 recall sessions, 294 attacks, 7 live sessions per cell) is well below the level-7 targets; live intervals are wide, and 7 paired sessions cannot reach significance for the live recall effects after correction.
2) **Scoring.** Recall, leaks and desynchronisation are scored automatically (key words, secret keywords, refusal and arrival cues), and the contradiction judge is an LLM. Blinded sheets for two raters (1,560 recall, 200 dialogue-quality and 133 contradiction items) are exported, but no human labels, Cohen's κ or judge agreement exist yet. The leak scorer over-counts: the two leaks flagged in the F4 probe context are W1 replies that use the secret keyword "looking" in "if you're looking for".
3) **Post-hoc corrections.** The retrieval scoring rule (facts older than the buffer) was chosen after the first Review-03 run

and applied unchanged, to every condition, in the reported run; the separation of the repair call from local compute is recomputed from the reported run's saved rows.

4) **Worlds.** W1 is the development world, W4 is templated, and W3 was authored by the notebook author rather than an independent contributor.
5) **Scripted players.** Fixed scripts remove adaptation confounds but do not react to NPCs; player experience needs a user study.
6) **Backbones.** Four self-hosted models of up to 7B parameters; the game's default hosted backbone was not evaluated, and its latency is not reproduced.
7) **Baselines.** B4 and B5 re-implement the retrieval rules of Generative Agents and MemoryBank, not the full systems; B4 omits the importance term and reflection.
8) **Before-and-after comparison.** F4v1 shares the corrected reducer, event store and commit node with F4, so it isolates the validator, prompt, packing and repair changes, not the recovery changes, which the crash trials measure separately.
9) **Violation metric.** Equation (6) counts flagged turns over transitions; persistent corruption inflates direct-write rates, so live violation rates compare conditions but are not counts of distinct violations.
10) **Memory scale.** Exact search with pruning at 1,000 entries; campaigns of thousands of turns need summarisation or hierarchical memory.

A. Threats to validity

Internal: inputs, prompts, schemas and backbones are identical across conditions, request seeds depend only on the probe or turn, and probe outputs are committed in all four modes, so control comparisons are exactly paired. *Construct*: violations are measured by an independent oracle written from the specification; recall is key-word matched on invented values; retrieval is scored on facts outside the buffer. *External*: claims are restricted to the tested worlds and models. *Conclusion*: hypotheses were stated in advance, the confirmatory family is Holm-corrected, unsupported hypotheses are reported, and the one scoring rule changed after the first Review-03 run (H2's) was fixed before the reported run. The factorial p-values are approximate posterior summaries from a variational fit.

XII. Conclusion

LLM NPCs need to remember what happened, respect the rules of the world, and do both at interactive cost. We tested an engine that pairs scoped dual-tier memory with a validated, event-sourced state against independent baselines at an equal memory budget, in a factorial design with ablations, on seven worlds and four models, with adversarial probes scored by an oracle that shares no code with the validator, and against its own earlier version. Three claims are supported. Long-term retrieval scoped to the current place and person keeps recall high and flat over 160 turns where a rolling context of equal size forgets everything after ten, and ranks the right memory higher than flat vector memory. A pre-commit barrier that checks each turn against a running state, over an event-sourced log, committed no state violation for any model while rejecting few lawful requests. Snapshot-plus-suffix recovery reproduced the committed state after every simulated crash. Three claims are not supported: the memory design is not clearly better than flat vector or forgetting-curve memory at an equal budget, memory and control are not independent, and one rewrite of a rejected reply did not reduce contradictions between dialogue and state.

A. Future work

Rerun at a larger sample size, which the planner can now reach within one Kaggle session; collect the two-rater human labels and validate the contradiction judge against them; author W3 independently; evaluate the hosted backbone; improve dialogue repair or generate narration from the committed events; ablate the packing rule's duplicate removal; probe recall away from the place a fact was told; validate narrative plausibility, not only structural legality; and evaluate filtered or hierarchical indices for long campaigns.

B. Data and licences

W2 content derives from LIGHT under CC BY-NC 4.0, so the released W2 worlds are for non-commercial use. The K2-Horizon backbones are Apache 2.0; the licences of the other two models are given on their model cards.

APPENDIX A
SUPPLEMENTARY RESULTS

This appendix holds results that the body states in text and that duplicate another table or figure.

Table XVII gives recall by fact age for every condition, the values plotted in Fig. 5.

TABLE XVII
RECALL (%) BY FACT AGE ON HELD-OUT WORLDS. AGES 80 AND 160 OCCUR ONLY IN 100- AND 200-TURN SESSIONS.

Condition	5	10	20	40	80	160
B0 Full history	77.8 [66.7, 87.5]	77.8 [69.4, 86.1]	81.9 [72.2, 90.3]	72.2 [62.5, 81.9]	79.2 [66.7, 89.6]	83.3 [70.8, 95.8]
B1 Rolling, B	79.2 [69.4, 88.9]	63.9 [51.4, 76.4]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]
B1 Rolling, $2B$	80.6 [72.2, 88.9]	73.6 [63.9, 81.9]	56.9 [44.4, 69.4]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]
B2 Vector memory	86.1 [77.8, 93.1]	87.5 [79.2, 94.4]	81.9 [72.2, 90.3]	79.2 [69.4, 88.9]	77.1 [64.6, 89.6]	91.7 [79.2, 100.0]
B3 Generic RAG	69.4 [59.7, 79.2]	65.3 [54.2, 76.4]	59.7 [48.6, 70.8]	56.9 [45.8, 68.1]	58.3 [45.8, 70.8]	62.5 [45.8, 79.2]
B4 Memory stream (GA)	87.5 [80.6, 94.4]	83.3 [73.6, 91.7]	81.9 [72.2, 90.3]	68.1 [58.3, 77.8]	35.4 [22.9, 47.9]	29.2 [12.5, 45.8]
B5 Forgetting curve	90.3 [83.3, 95.8]	88.9 [80.6, 95.8]	81.9 [73.6, 88.9]	80.6 [72.2, 88.9]	66.7 [52.1, 79.2]	87.5 [75.0, 100.0]
F4v1 Review-02 system	95.8 [90.3, 100.0]	90.3 [81.9, 97.2]	81.9 [72.2, 90.3]	86.1 [79.2, 93.1]	81.2 [70.8, 89.6]	95.8 [87.5, 100.0]
F4 Proposed	81.9 [73.6, 90.3]	80.6 [70.8, 90.3]	83.3 [75.0, 91.7]	79.2 [69.4, 87.5]	81.2 [68.8, 93.8]	91.7 [79.2, 100.0]
A1 No long-term memory	88.9 [81.9, 95.8]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]
A2 No short-term buffer	86.1 [77.8, 93.1]	88.9 [81.9, 95.8]	84.7 [76.4, 91.7]	84.7 [75.0, 93.1]	81.2 [70.8, 89.6]	87.5 [75.0, 100.0]
A3 No scoping	72.2 [62.5, 81.9]	84.7 [76.4, 91.7]	86.1 [76.4, 94.4]	72.2 [62.5, 81.9]	77.1 [64.6, 87.5]	87.5 [75.0, 100.0]
A4 No fallback	87.5 [80.6, 94.4]	79.2 [69.4, 88.9]	83.3 [75.0, 91.7]	79.2 [69.4, 87.5]	81.2 [68.8, 93.8]	91.7 [79.2, 100.0]
n (F4 probes)	72	72	72	72	48	24

Table XVIII lists retrieval quality and secondary memory metrics for every memory condition.

TABLE XVIII
RETRIEVAL QUALITY AND SECONDARY MEMORY METRICS (HELD-OUT WORLDS). RECALL@6 IS SCORED ON THE 288 PROBES OLDER THAN THE BUFFER; THE OTHER COLUMNS ON ALL 360. "CONFUSED": ANSWERED WITH THE DISTRACTOR'S VALUE. NO PROMPT REACHED THE 16,384-CHARACTER CAP.

Condition	Gold in prompt	Distractor in prompt	Confused	Recall@6	Parse success	In / out tok. per 100 turns
F4	100.0 [100.0, 100.0]	55.3 [50.5, 60.4]	2.2 [0.8, 3.9]	100.0 [100.0, 100.0]	96.1 [93.5, 98.3]	144,512 / 14,866
F4v1	100.0 [100.0, 100.0]	57.2 [52.2, 62.6]	0.8 [0.0, 1.8]	100.0 [100.0, 100.0]	97.2 [95.2, 98.9]	140,045 / 15,388
B0	100.0 [100.0, 100.0]	100.0 [100.0, 100.0]	4.7 [2.8, 6.8]	–	98.6 [97.4, 99.7]	441,351 / 13,578
B1, B	34.4 [30.7, 38.3]	57.5 [53.2, 62.1]	12.5 [8.4, 16.8]	–	99.7 [99.1, 100.0]	147,773 / 14,124
B1, $2B$	54.4 [50.0, 59.2]	73.6 [69.7, 77.7]	11.7 [8.8, 14.8]	–	98.6 [97.4, 99.7]	187,553 / 13,357
B2	100.0 [100.0, 100.0]	98.9 [97.5, 100.0]	0.3 [0.0, 0.8]	100.0 [100.0, 100.0]	98.9 [97.8, 99.7]	145,077 / 15,184
B3	100.0 [100.0, 100.0]	94.4 [91.5, 96.9]	10.0 [7.6, 12.4]	100.0 [100.0, 100.0]	98.1 [96.3, 99.4]	150,573 / 14,925
B4	86.9 [84.5, 89.4]	91.7 [89.3, 93.9]	1.9 [0.8, 3.3]	83.7 [80.7, 86.7]	96.9 [95.2, 98.6]	144,938 / 15,386
B5	96.4 [94.4, 98.3]	90.6 [87.3, 93.8]	0.8 [0.0, 1.9]	95.5 [93.1, 97.8]	98.3 [97.1, 99.4]	145,012 / 15,272
A1	20.0 [19.0, 21.1]	52.5 [48.1, 57.3]	15.8 [11.6, 20.4]	–	98.6 [97.1, 99.7]	140,398 / 14,523
A2	100.0 [100.0, 100.0]	11.4 [8.2, 15.0]	0.0 [0.0, 0.0]	100.0 [100.0, 100.0]	99.4 [98.6, 100.0]	118,208 / 14,384
A3	98.9 [97.8, 99.7]	90.8 [88.1, 93.5]	1.9 [0.6, 3.5]	98.6 [97.2, 99.7]	99.4 [98.6, 100.0]	144,391 / 14,724
A4	100.0 [100.0, 100.0]	50.6 [46.4, 55.1]	1.7 [0.3, 3.3]	100.0 [100.0, 100.0]	95.8 [93.3, 98.1]	144,221 / 14,946

Fig. 8 plots the retrieval metrics and the gold-in-prompt rate by fact age.

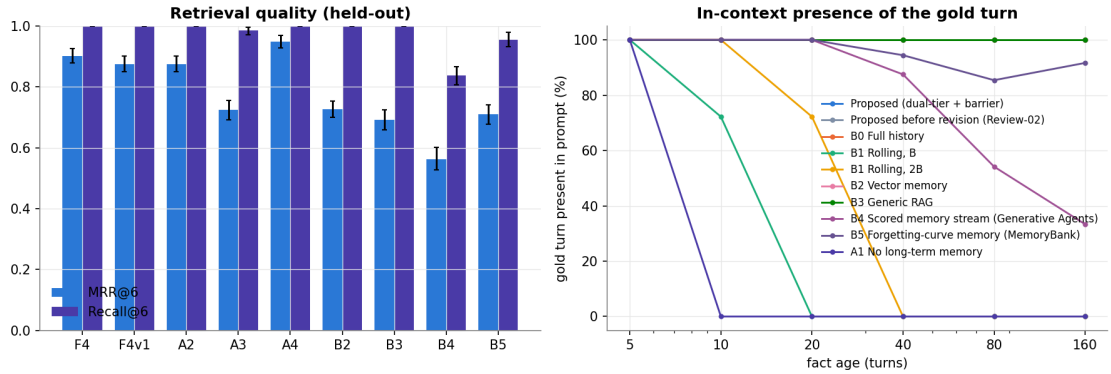


Fig. 8. Left: MRR@6 and Recall@6 of the gold memory for each retrieval condition, on the 288 held-out probes older than the buffer. Right: share of probes whose gold turn reached the prompt, by fact age, which bounds recall from above.

Table XIX lists the sensitivity sweeps plotted in Fig. 6.

TABLE XIX
SENSITIVITY SWEEPS (RETRIEVAL ONLY; W1 AND W2A TO W2C, HORIZONS 100 AND 200; 880 PROBES PER SETTING).

	buffer 4	buffer 8	buffer 16	$k = 3$	$k = 12$	$\theta = 0$	$\theta = 6$
Gold in prompt	0.9989	0.9989	0.9989	0.9648	1.0000	1.0000	0.9955
MRR@k	0.8990	0.8990	0.8990	0.8390	0.9124	0.9227	0.8303

Fig. 9 plots the budget sweep and the trade-off between recall, tokens and latency.

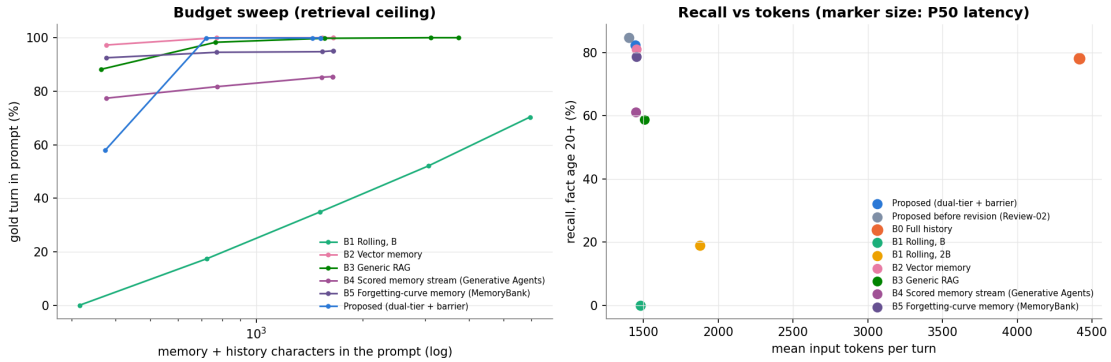


Fig. 9. Left: share of probes whose gold turn reaches the prompt as each condition's budget is scaled from 0.25 to 4 times 1,600 characters (CPU sweep on W1 and W2). Right: recall at fact age 20 or more against mean input tokens, with marker size proportional to P50 single-stream latency, which is between 6.3 and 7.4 s for every condition except B0.

Fig. 10 breaks the live violations of Table VIII down by invariant.

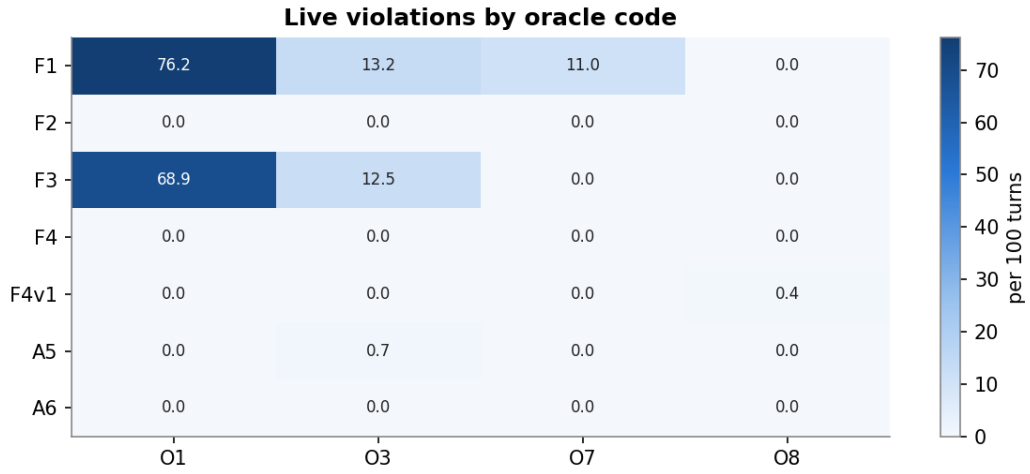


Fig. 10. Oracle-flagged live turns per 100 turns by condition and invariant code (codes that never fired are omitted by the plot). Direct writes fail on O1 and O3 (F1 and F3) and O7 (F1); the corrected-barrier conditions F2 and F4, and A6, have none; A5 has 0.7 O3 and F4v1 0.4 O8 per 100 turns.

Fig. 11 plots the main rows of Table XII.

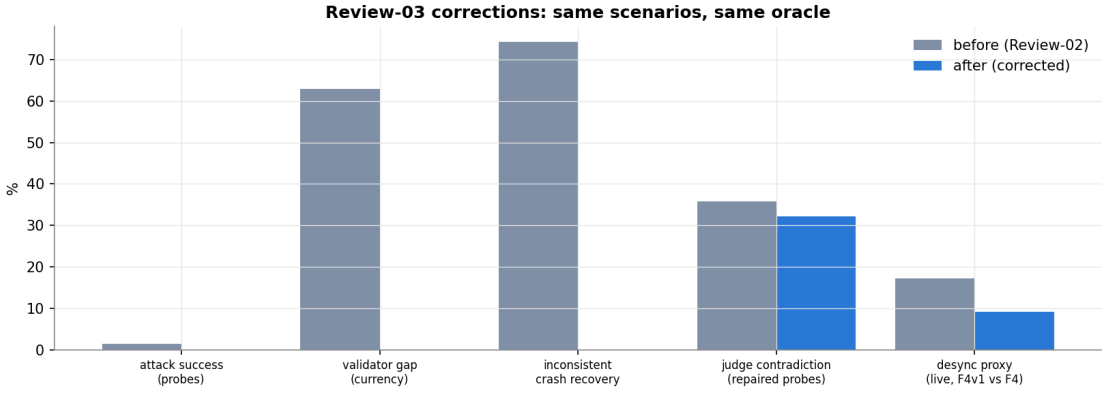


Fig. 11. Before (Review-02 system, grey) and after (corrected system, blue) on the same scenarios: attack success and currency validator gap under the two validators, inconsistent crash recoveries under the two load paths, judged contradictions of draft and repaired probe replies, and the live desynchronisation proxy of F4v1 and F4.

Fig. 12 plots the seed variance of Table VI and the recall of all four models on their common subset.

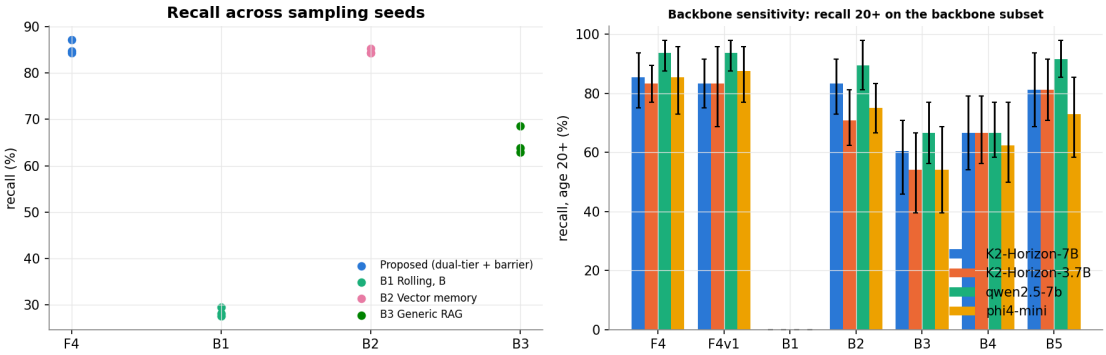


Fig. 12. Left: recall over all ages for three sampling seeds on the calibration sessions. Right: recall at fact age 20 or more on the common subset (W1, W2a, W3, W4-32 at horizon 100; 48 probes per bar) for K2-Horizon-7B, K2-Horizon-3.7B, Qwen2.5-7B and Phi-4-mini; F4 stays between 83% and 94% across models, while B2 and B3 vary more.

Fig. 13 plots the attack success and validator errors of Table VII.

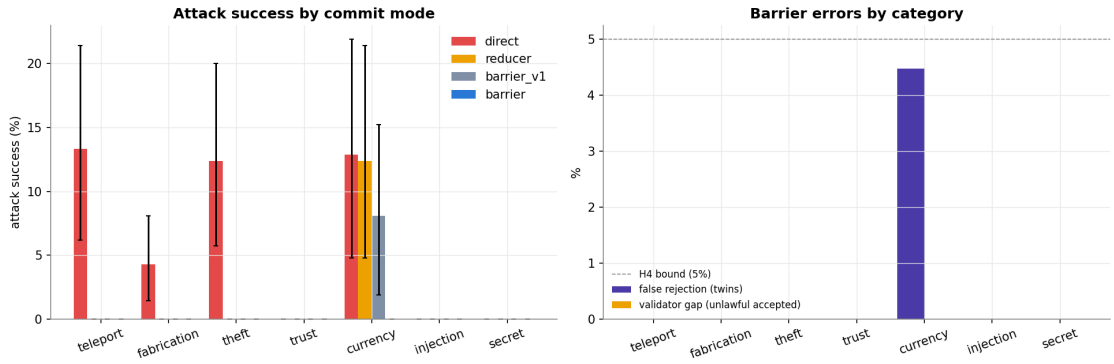


Fig. 13. Left: attack success by category when the same model outputs are committed by direct writes, by the reducer without validation (A5), through the Review-02 barrier and through the corrected barrier. Right: the corrected barrier's false rejection of lawful twins (three currency twin proposals) and its validator gap (zero in every category), against the 5% bound of H4.

Fig. 14 plots the 2×2 cells of Table VIII.

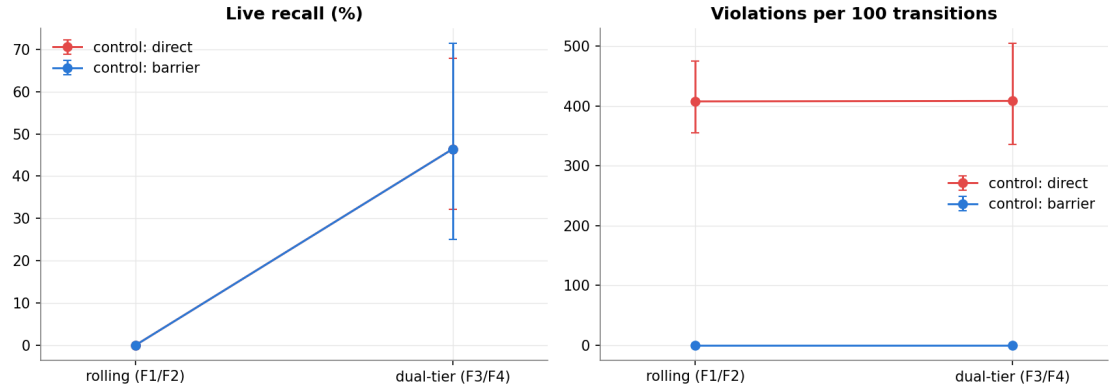


Fig. 14. Interaction plots for the 2×2 design. Left: live recall rises with memory under either control setting and does not change with the barrier. Right: violations per 100 transitions fall to zero with the barrier under either memory setting. The mixed model (Table IX) nevertheless finds a recall interaction.

Table XX gives the per-stage latency of the live F4 graph.

TABLE XX
PER-STAGE LATENCY (MS) OF THE LIVE F4 GRAPH, SINGLE STREAM (24 UNCACHED TURNS, TWO WORLDS). REPAIR IS THE DIALOGUE-REPAIR LLM CALL, NEEDED ON 2 OF THE 24 TURNS; LOCAL TOTAL EXCLUDES THE LLM AND REPAIR CALLS. THE REPAIR AND LOCAL COLUMNS ARE RECOMPUTED FROM METRICS/LATENCY_SINGLE_STREAM_LIVE.CSV (SECTION VIII-E).

	Input	Retrieval	Prompt	LLM	Parse	Validate	Repair	Commit	Output	Local total	End to end
P50	0.003	0.172	0.146	8,839	0.090	0.014	0.002	9.726	0.012	18.192	8,855
P95	0.003	0.229	0.188	16,101	0.106	1.365	8,918	15.389	0.014	23.435	24,125
P99	0.003	0.236	0.199	16,697	0.110	1.387	11,189	16.455	0.015	24.367	26,921

Fig. 15 plots the stage latencies of Table XX and the single-stream LLM time per condition.

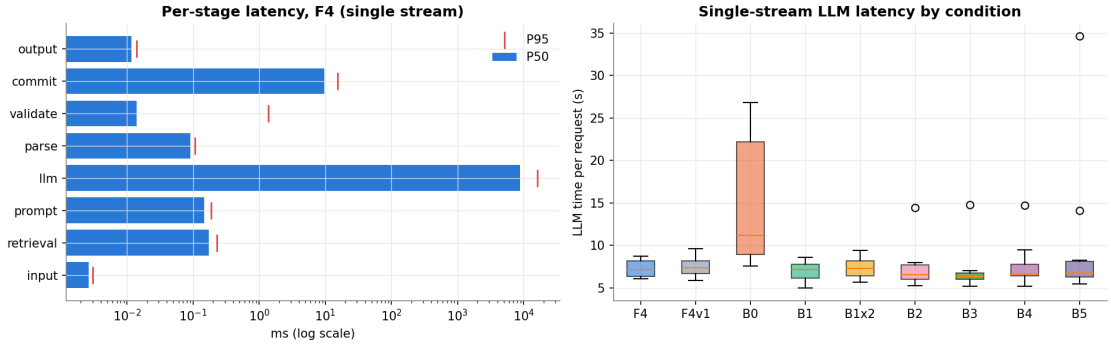


Fig. 15. Left: P50 (bars) and P95 (ticks) latency of each F4 stage, single stream, log scale, as plotted by the notebook, which has no repair stage (Table XX). Right: single-stream LLM time per request for each condition's recall prompts (12 requests per condition).

Fig. 16 plots the per-world effects of Table XIII.

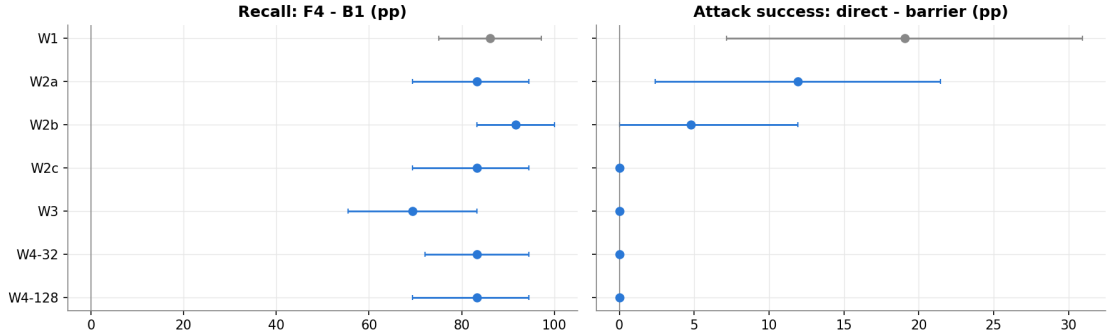


Fig. 16. Per-world effects with 95% bootstrap intervals. Left: recall gain of F4 over B1 at fact age 20 or more, positive on every world. Right: reduction in attack success from direct writes to the barrier (F4 context); positive on W1, W2a and W2b and zero on W2c, W3, W4-32 and W4-128, where no attack succeeded under direct writes. W1 (grey) is in-distribution.

APPENDIX B
SAMPLE-SIZE LADDER

Table XXI lists the eight pre-declared sample-size levels of Section VIII-B.

TABLE XXI
PRE-DECLARED SAMPLE-SIZE LEVELS AND THE PLANNER'S ESTIMATED HOURS, INCLUDING THE 1.15 SAFETY FACTOR THAT THE RELEASED NOTEBOOK NO LONGER APPLIES (METRICS/BUDGET_PLAN.CSV). LEVEL 1 WAS RUN.

Level	Sessions/(world, H)	Probes/(world, cat.)	Live/world	Recall (h)	Probes (h)	Live (h)	Seed var. (h)	Total (h)
0	1	4	1	0.93	0.62	0.91	0.56	3.49
1	**2**	**6**	**1**	**1.93**	**0.97**	**0.91**	**0.56**	**5.03**
2	3	10	2	2.93	1.67	1.82	0.56	8.03
3	4	14	2	3.92	2.37	1.82	0.56	9.98
4	6	20	3	5.91	3.42	2.73	0.56	14.52
5	10	30	4	9.90	5.17	3.64	0.56	22.16
6	20	45	6	19.85	7.79	4.55	0.56	37.66
7	40	60	10	39.76	10.41	7.28	0.56	66.72

REFERENCES

[1] R. Gallotta, G. Todd, M. Zammit, S. Earle, A. Liapis, J. Togelius, and G. N. Yannakakis, "Large language models and games: A survey and roadmap," *IEEE Transactions on Games*, 2024. [Online]. Available: https://doi.org/10.1109/TG.2024.3461510

[2] N. F. Liu, K. Lin, J. Hewitt, A. Paranjape, M. Bevilacqua, F. Petroni, and P. Liang, "Lost in the middle: How language models use long contexts," *Transactions of the Association for Computational Linguistics*, vol. 12, pp. 157–173, 2024. [Online]. Available: https://doi.org/10.1162/tacl_a_00638

[3] A. Maharana, D.-H. Lee, S. Tulyakov, M. Bansal, F. Barbieri, and Y. Fang, "Evaluating very long-term conversational memory of LLM agents," in *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 2024, pp. 13 851–13 870. [Online]. Available: https://doi.org/10.18653/v1/2024.acl-long.747

[4] D. Wu, H. Wang, W. Yu, Y. Zhang, K.-W. Chang, and D. Yu, "LongMemEval: Benchmarking chat assistants on long-term interactive memory," in *The Thirteenth International Conference on Learning Representations (ICLR)*, 2025. [Online]. Available: https://proceedings.iclr.cc/paper_files/paper/2025/hash/d813d324dbf0598bbdc9c8e79740ed01-Abstract-Conference.html

[5] R. Wang, G. Todd, Z. Xiao, X. Yuan, M.-A. Côté, P. Clark, and P. Jansen, "Can language models serve as text-based world simulators?" in *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*, 2024, pp. 1–17. [Online]. Available: https://doi.org/10.18653/v1/2024.acl-short.1

[6] N. Kim and S. Schuster, "Entity tracking in language models," in *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 2023, pp. 3835–3855. [Online]. Available: https://doi.org/10.18653/v1/2023.acl-long.213

[7] Y. Ma, J. Yan, B. Shi, K. H. Kam, R. Wang, X. Liu, Y. Chen, Y. Zhang, and D. F. Wong, "Can LLM agents stick to the script? A benchmark for long-horizon consistency in interactive narratives," in *Proceedings of the 43rd International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 306, 2026, pp. 85 171–85 192. [Online]. Available: https://proceedings.mlr.press/v306/ma26ap.html

[8] J. S. Park, J. C. O'Brien, C. J. Cai, M. R. Morris, P. Liang, and M. S. Bernstein, "Generative agents: Interactive simulacra of human behavior," in *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology (UIST)*, 2023, pp. 1–22. [Online]. Available: https://doi.org/10.1145/3586183.3606763

[9] N. Shinn, F. Cassano, A. Gopinath, K. Narasimhan, and S. Yao, "Reflexion: Language agents with verbal reinforcement learning," in *Advances in Neural Information Processing Systems*, vol. 36, 2023. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2023/hash/1b44b878bb782e6954cd888628510e90-Abstract-Conference.html

[10] J. Urbanek, A. Fan, S. Karamcheti, S. Jain, S. Humeau, E. Dinan, T. Rocktäschel, D. Kiela, A. Szlam, and J. Weston, "Learning to speak and act in a fantasy text adventure game," in *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, 2019, pp. 673–683. [Online]. Available: https://doi.org/10.18653/v1/D19-1062

[11] P. Ammanabrolu, J. Urbanek, M. Li, A. Szlam, T. Rocktäschel, and J. Weston, "How to motivate your dragon: Teaching goal-driven agents to speak and act in fantasy worlds," in *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL-HLT)*, 2021, pp. 807–833. [Online]. Available: https://doi.org/10.18653/v1/2021.naacl-main.64

[12] M.-A. Côté, Á. Kádár, X. Yuan, B. Kybartas, T. Barnes, E. Fine, J. Moore, M. Hausknecht, L. El Asri, M. Adada, W. Tay, and A. Trischler, "TextWorld: A learning environment for text-based games," in *Computer Games (CGW 2018)*, ser. Communications in Computer and Information Science, vol. 1017. Springer, 2019, pp. 41–75. [Online]. Available: https://doi.org/10.1007/978-3-030-24337-1_3

[13] M. Hausknecht, P. Ammanabrolu, M.-A. Côté, and X. Yuan, "Interactive fiction games: A colossal adventure," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 34, 2020, pp. 7903–7910. [Online]. Available: https://doi.org/10.1609/aaai.v34i05.6297

[14] C. Callison-Burch, G. Singh Tomar, L. J. Martin, D. Ippolito, S. Bailis, and D. Reitter, "Dungeons and dragons as a dialog challenge for artificial intelligence," in *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2022, pp. 9379–9393. [Online]. Available: https://doi.org/10.18653/v1/2022.emnlp-main.637

[15] N. Akoury, S. Wang, J. Whiting, S. Hood, N. Peng, and M. Iyyer, "STORIUM: A dataset and evaluation platform for machine-in-the-loop story generation," in *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2020, pp. 6470–6484. [Online]. Available: https://doi.org/10.18653/v1/2020.emnlp-main.525

[16] J. Xu, A. Szlam, and J. Weston, "Beyond goldfish memory: Long-term open-domain conversation," in *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 2022, pp. 5180–5197. [Online]. Available: https://doi.org/10.18653/v1/2022.acl-long.356

[17] W. Zhong, L. Guo, Q. Gao, H. Ye, and Y. Wang, "MemoryBank: Enhancing large language models with long-term memory," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 38, 2024, pp. 19 724–19 731. [Online]. Available: https://doi.org/10.1609/aaai.v38i17.29946

[18] B. Jiménez Gutiérrez, Y. Shu, Y. Gu, M. Yasunaga, and Y. Su, "HippoRAG: Neurobiologically inspired long-term memory for large language models," in *Advances in Neural Information Processing Systems*, vol. 37, 2024. [Online]. Available: https://doi.org/10.52202/079017-1902

[19] W. Xu, Z. Liang, K. Mei, H. Gao, J. Tan, and Y. Zhang, "A-Mem: Agentic memory for LLM agents," in *Advances in Neural Information Processing Systems*, vol. 38, 2025. [Online]. Available: https://doi.org/10.52202/085713-0593

[20] Z. Tan, J. Yan, I.-H. Hsu, R. Han, Z. Wang, L. Le, Y. Song, Y. Chen, H. Palangi, G. Lee, A. R. Iyer, T. Chen, H. Liu, C.-Y. Lee, and T. Pfister, "In prospect and retrospect: Reflective memory management for long-term personalized dialogue agents," in *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 2025, pp. 8416–8439. [Online]. Available: https://doi.org/10.18653/v1/2025.acl-long.413

[21] J. Liu, Y. Su, P. Xia, S. Han, Z. Zheng, C. Xie, M. Ding, and H. Yao, "SimpleMem: Efficient lifelong memory for LLM agents," in *Proceedings of the 43rd International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 306, 2026, pp. 78 500–78 513. [Online]. Available: https://proceedings.mlr.press/v306/liu26dx.html

[22] T. R. Sumers, S. Yao, K. Narasimhan, and T. L. Griffiths, "Cognitive architectures for language agents," *Transactions on Machine Learning Research*, 2024. [Online]. Available: https://openreview.net/forum?id=1i6ZCvflQJ

[23] Z. Zhang, Q. Dai, X. Bo, C. Ma, R. Li, X. Chen, J. Zhu, Z. Dong, and J.-R. Wen, "A survey on the memory mechanism of large language model-based agents," *ACM Transactions on Information Systems*, vol. 43, no. 6, pp. 1–47, 2025. [Online]. Available: https://doi.org/10.1145/3748302

[24] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Küttler, M. Lewis, W.-t. Yih, T. Rocktäschel, S. Riedel, and D. Kiela, "Retrieval-augmented generation for knowledge-intensive NLP tasks," in *Advances in Neural Information Processing Systems*, vol. 33, 2020, pp. 9459–9474. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2020/hash/6b493230205f780e1bc26945df7481e5-Abstract.html

[25] S. Yao, J. Zhao, D. Yu, N. Du, I. Shafran, K. Narasimhan, and Y. Cao, "ReAct: Synergizing reasoning and acting in language models," in *The Eleventh International Conference on Learning Representations (ICLR)*, 2023. [Online]. Available: https://openreview.net/forum?id=WE_vluYUL-X

[26] S. Gollapudi, N. Karia, V. Sivashankar, R. Krishnaswamy, N. Begwani, S. Raz, Y. Lin, Y. Zhang, N. Mahapatro, P. Srinivasan, A. Singh, and H. V. Simhadri, "Filtered-DiskANN: Graph algorithms for approximate nearest neighbor search with filters," in *Proceedings of the ACM Web Conference 2023 (WWW)*, 2023, pp. 3406–3416. [Online]. Available: https://doi.org/10.1145/3543507.3583552

[27] S. Kambhampati, K. Valmeekam, L. Guan, M. Verma, K. Stechly, S. Bhambri, L. P. Saldyt, and A. B. Murthy, "Position: LLMs can't plan, but can help planning in LLM-modulo frameworks," in *Proceedings of the 41st International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 235, 2024, pp. 22 895–22 907. [Online]. Available: https://proceedings.mlr.press/v235/kambhampati24a.html

[28] S. Geng, M. Josifoski, M. Peyrard, and R. West, "Grammar-constrained decoding for structured NLP tasks without finetuning," in *Proceedings of the 2023 Conference on Empirical Methods in Natural Language*

Processing (EMNLP), 2023, pp. 10 932–10 952. [Online]. Available: https://doi.org/10.18653/v1/2023.emnlp-main.674

[29] K. Greshake, S. Abdelnabi, S. Mishra, C. Endres, T. Holz, and M. Fritz, "Not what you've signed up for: Compromising real-world LLM-integrated applications with indirect prompt injection," in *Proceedings of the 16th ACM Workshop on Artificial Intelligence and Security (AISec)*, 2023, pp. 79–90. [Online]. Available: https://doi.org/10.1145/3605764.3623985

[30] M. Overeem, M. Spoor, S. Jansen, and S. Brinkkemper, "An empirical characterization of event sourced systems and their schema evolution: Lessons from industry," *Journal of Systems and Software*, vol. 178, p. 110970, 2021. [Online]. Available: https://doi.org/10.1016/j.jss.2021.110970

[31] S. Buongiorno, L. Klinkert, Z. Zhuang, T. Chawla, and C. Clark, "PANGeA: Procedural artificial narrative using generative AI for turn-based, role-playing video games," in *Proceedings of the AAAI Conference on Artificial Intelligence and Interactive Digital Entertainment (AIIDE)*, vol. 20, 2024, pp. 156–166. [Online]. Available: https://doi.org/10.1609/aiide.v20i1.31876

[32] A. Phillips, J. Lang, and D. Mould, "Goal-oriented interactions in games using LLMs," *IEEE Transactions on Games*, vol. 17, no. 2, pp. 510–521, 2025. [Online]. Available: https://doi.org/10.1109/TG.2024.3515807

[33] V. Kumaran, J. Rowe, B. Mott, and J. Lester, "SceneCraft: Automating interactive narrative scene generation in digital games with large language models," in *Proceedings of the AAAI Conference on Artificial Intelligence and Interactive Digital Entertainment (AIIDE)*, vol. 19, no. 1, 2023, pp. 86–96. [Online]. Available: https://doi.org/10.1609/aiide.v19i1.27504

[34] A. Zhu, L. Martin, A. Head, and C. Callison-Burch, "CALYPSO: LLMs as dungeon master's assistants," in *Proceedings of the AAAI Conference on Artificial Intelligence and Interactive Digital Entertainment (AIIDE)*, vol. 19, no. 1, 2023, pp. 380–390. [Online]. Available: https://doi.org/10.1609/aiide.v19i1.27534

[35] N. Reimers and I. Gurevych, "Sentence-BERT: Sentence embeddings using Siamese BERT-networks," in *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, 2019, pp. 3980–3990. [Online]. Available: https://doi.org/10.18653/v1/D19-1410

[36] W. Wang, F. Wei, L. Dong, H. Bao, N. Yang, and M. Zhou, "MiniLM: Deep self-attention distillation for task-agnostic compression of pre-trained transformers," in *Advances in Neural Information Processing Systems*, vol. 33, 2020, pp. 5776–5788. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2020/hash/3f5ee243547dee91fbd053c1c4a845aa-Abstract.html

[37] J. Johnson, M. Douze, and H. Jégou, "Billion-scale similarity search with GPUs," *IEEE Transactions on Big Data*, vol. 7, no. 3, pp. 535–547, 2021. [Online]. Available: https://doi.org/10.1109/TBDATA.2019.2921572

[38] B. Efron, "Bootstrap methods: Another look at the jackknife," *The Annals of Statistics*, vol. 7, no. 1, pp. 1–26, 1979. [Online]. Available: https://doi.org/10.1214/aos/1176344552

[39] C. A. Field and A. H. Welsh, "Bootstrapping clustered data," *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, vol. 69, no. 3, pp. 369–390, 2007. [Online]. Available: https://doi.org/10.1111/j.1467-9868.2007.00593.x

[40] Q. McNemar, "Note on the sampling error of the difference between correlated proportions or percentages," *Psychometrika*, vol. 12, no. 2, pp. 153–157, 1947. [Online]. Available: https://doi.org/10.1007/BF02295996

[41] F. Wilcoxon, "Individual comparisons by ranking methods," *Biometrics Bulletin*, vol. 1, no. 6, pp. 80–83, 1945. [Online]. Available: https://doi.org/10.2307/3001968

[42] S. Holm, "A simple sequentially rejective multiple test procedure," *Scandinavian Journal of Statistics*, vol. 6, no. 2, pp. 65–70, 1979. [Online]. Available: https://www.jstor.org/stable/4615733

[43] S. Seabold and J. Perktold, "Statsmodels: Econometric and statistical modeling with Python," in *Proceedings of the 9th Python in Science Conference*, 2010, pp. 92–96. [Online]. Available: https://doi.org/10.25080/Majora-92bf1922-011

[44] W. Kwon, Z. Li, S. Zhuang, Y. Sheng, L. Zheng, C. H. Yu, J. E. Gonzalez, H. Zhang, and I. Stoica, "Efficient memory management for large language model serving with PagedAttention," in *Proceedings of the 29th Symposium on Operating Systems Principles (SOSP)*, 2023, pp. 611–626. [Online]. Available: https://doi.org/10.1145/3600006.3613165